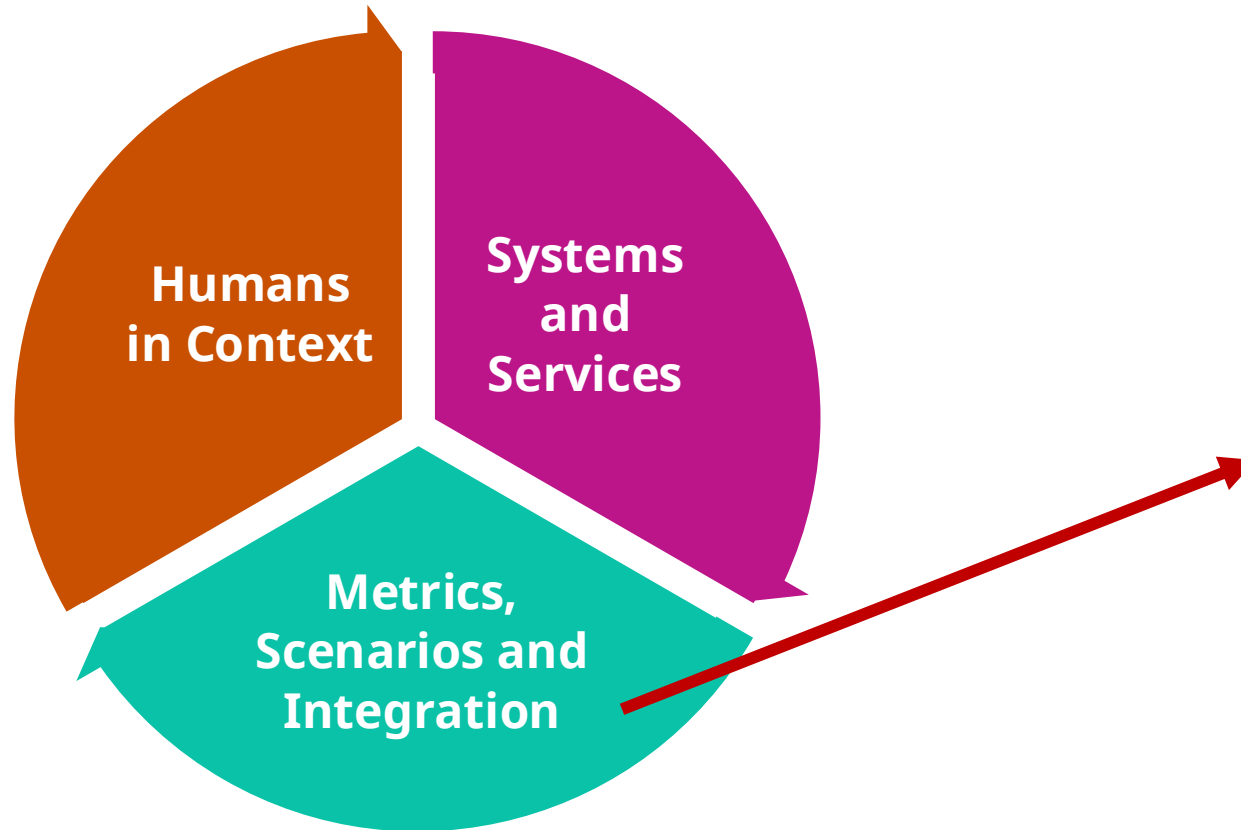


Synergy between aggregated anonymized real-world data and agent-based modeling: A case study of the Carola Bridge collapse in Dresden

The AgiMo project



More details on this:
session 105-18 by Daniel Röder

Collaborative Research Center SFB/TRR 408 “AgiMo:
Data-driven agile planning for responsible mobility”

The collapse of the Carola Bridge in Dresden



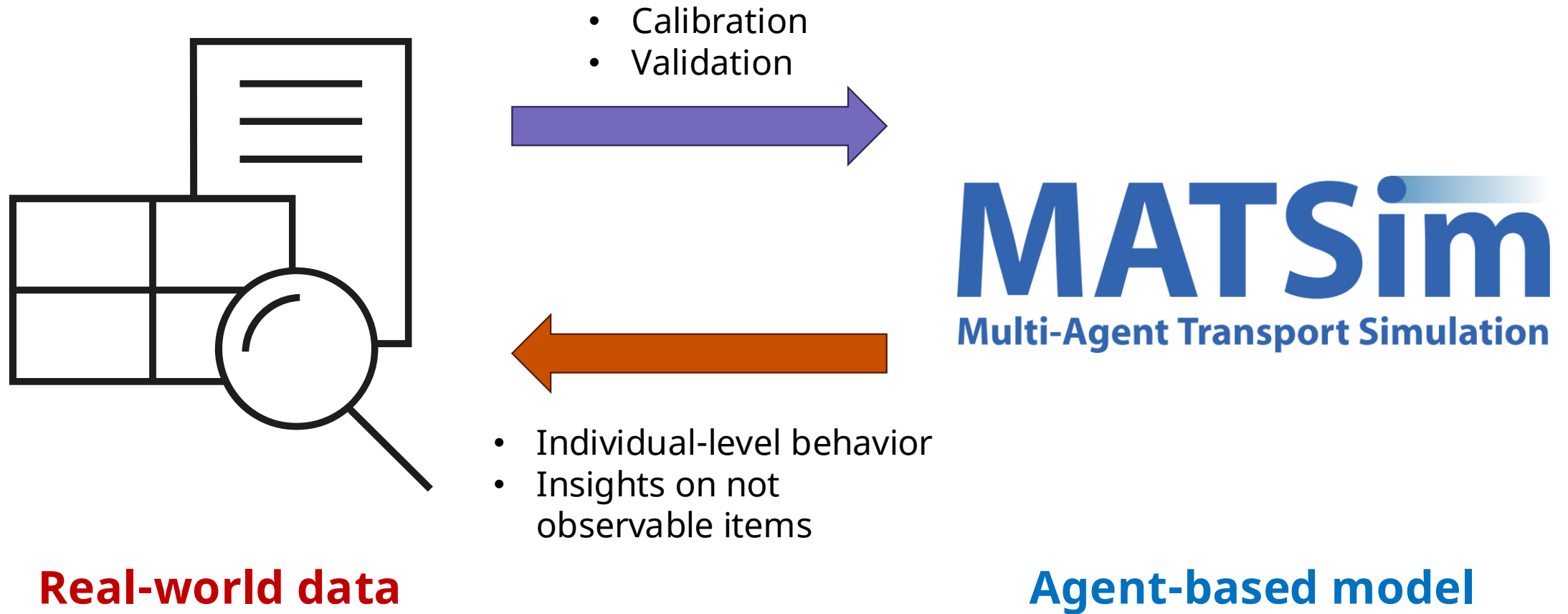
© Stefan Gröschel

On 11 September 2024, part of the Carola Bridge in Dresden collapsed.

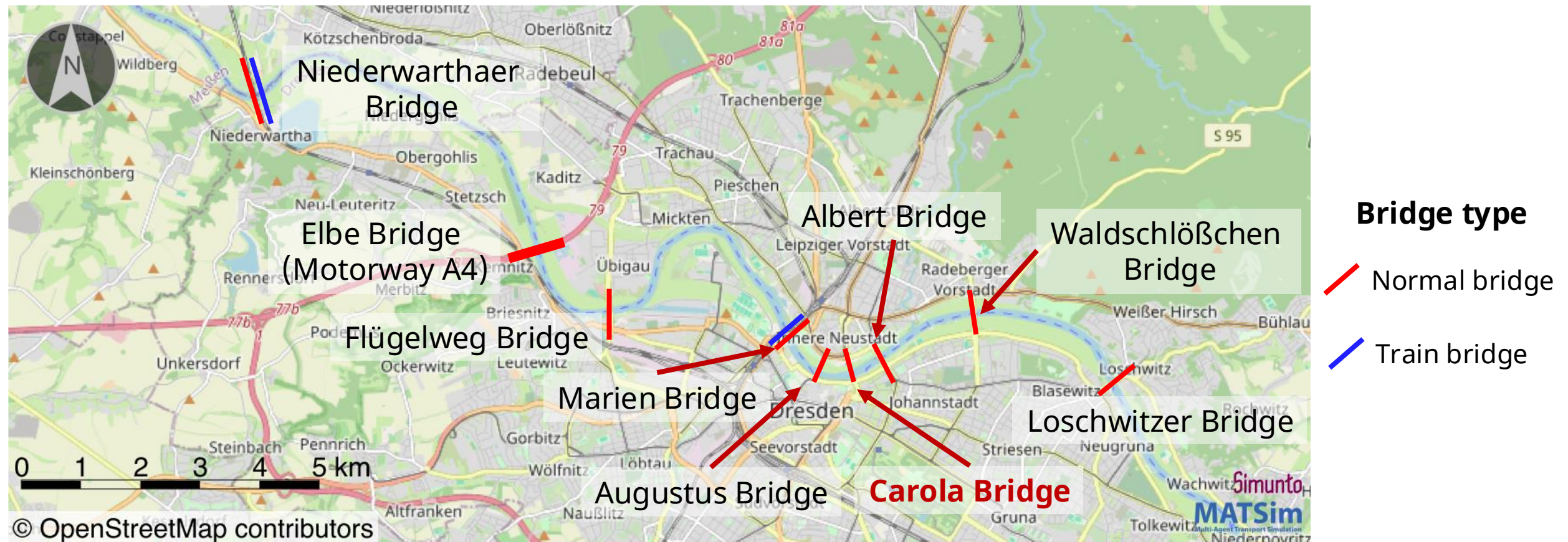
The rest part of the bridge is then closed and deconstructed.

The bridge was an arterial road in the city (Bundesstrasse, B170)

Synergy between real-world data and MATSim



The bridges in Dresden



Available real-world data

Traffic count of motorized vehicles on the bridges before and after the collapse of the Carola bridge

Passenger flow of the buses and trams crossing the Elbe river in Dresden

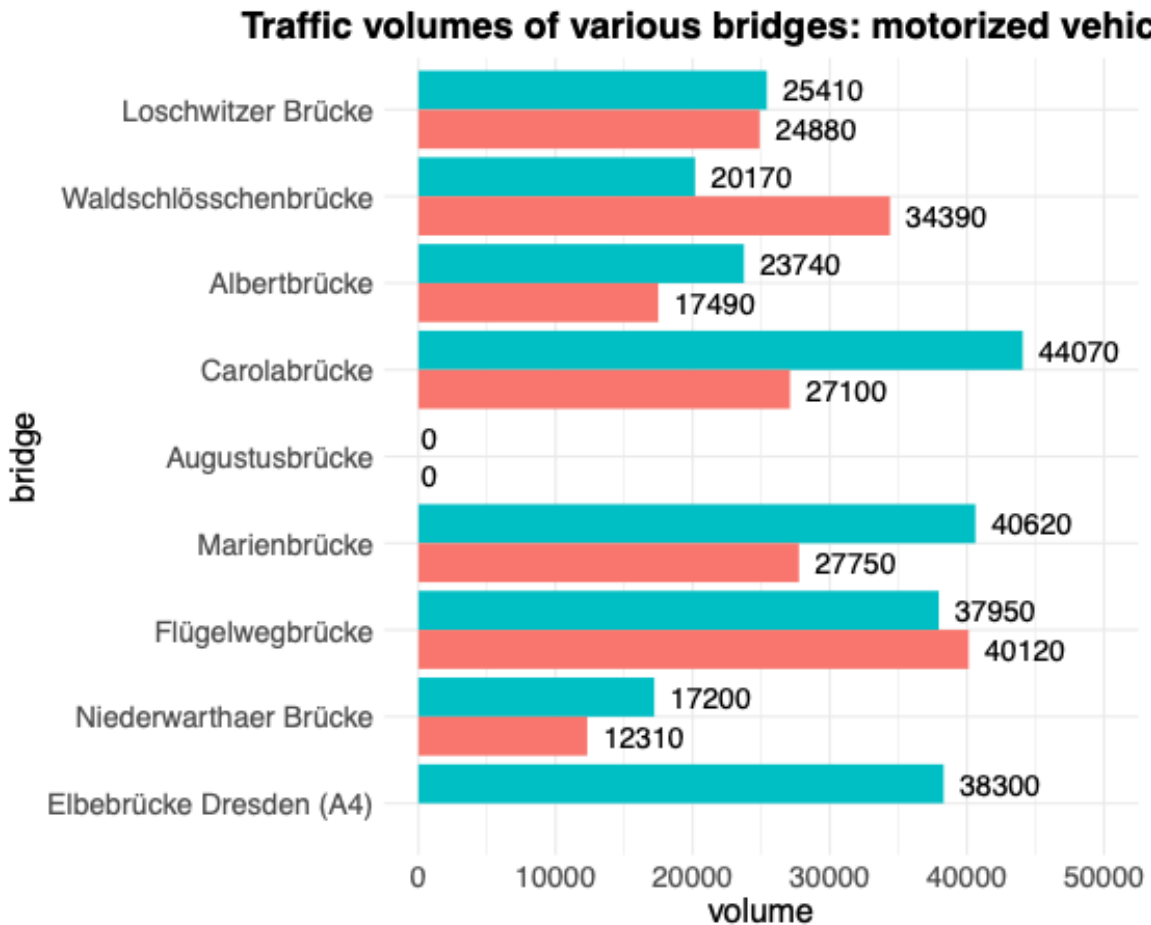
Automated bicycle count data

Motivation

1. Explore the behavior change after the collapse of the Carola bridge in the agent-based model
2. Filling the gap that real-world data does not cover
3. Perform agent-level analysis with the agent-based model

Matching the traffic count data on the bridges

The Dresden model v1.1



source
actual count 2023
simulated

Too much traffic on the Carola bridge

Total bridge crossing trips
(exclude A4)

Simulated: 209160

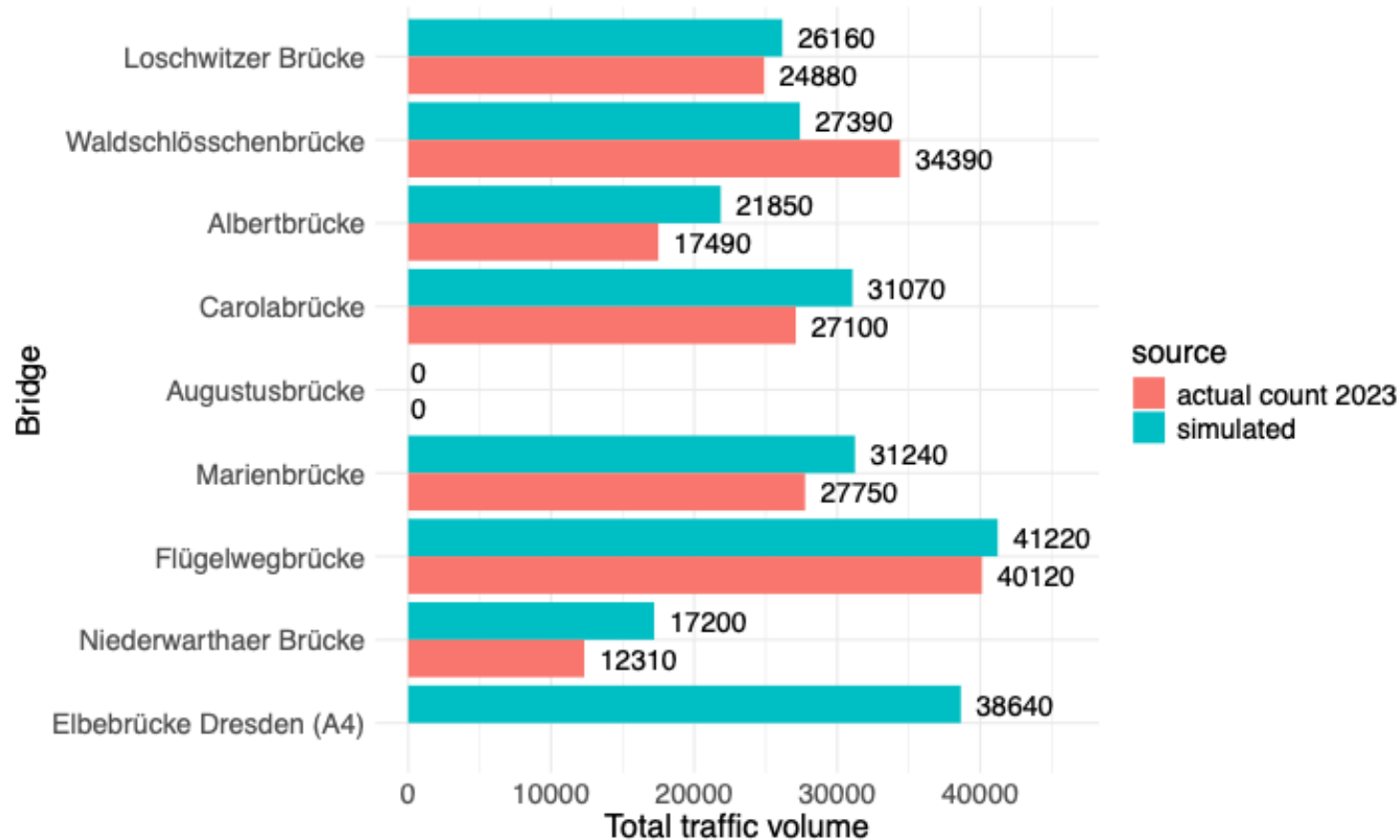
Actual: 184040

Difference: +13.6%

Matching the traffic count data on the bridges

The Dresden model v1.1 with modified network

Traffic volumes of various bridges: motorized vehicle (Kfz)



Evaluate and update the capacity of the bridges.

Reduce capacity of:

Carola bridge

Marien bridge

Albert bridge

Increase the capacity and freespeed of:

Waldschlosschen bridge

Total bridge crossing trips
(exclude A4)

Simulated: 196130

Actual: 184040

Difference: +6.6%

PT passenger traffic

Validation

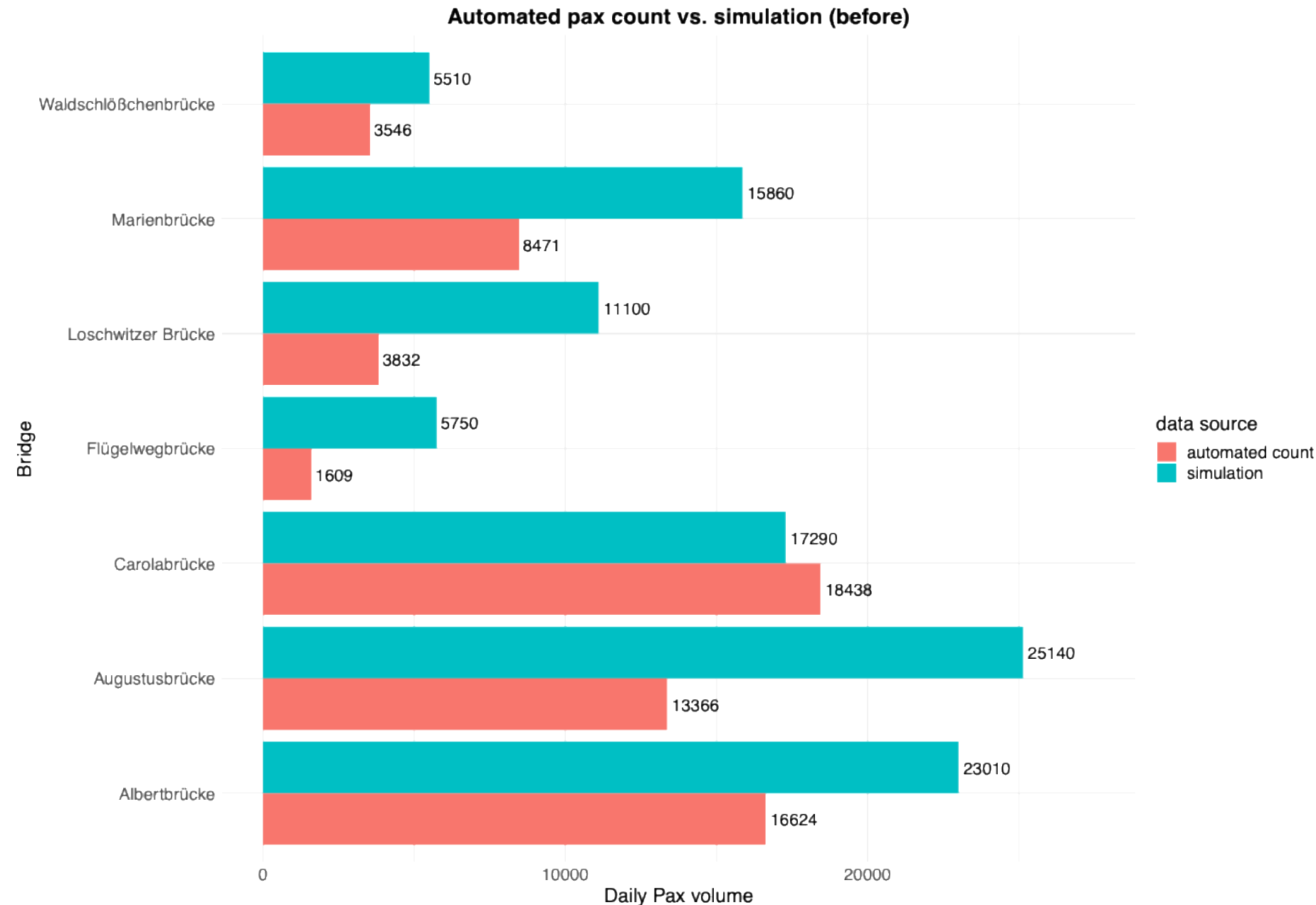
Before the collapse of the Carola bridge.

Total PAX on all bridges*

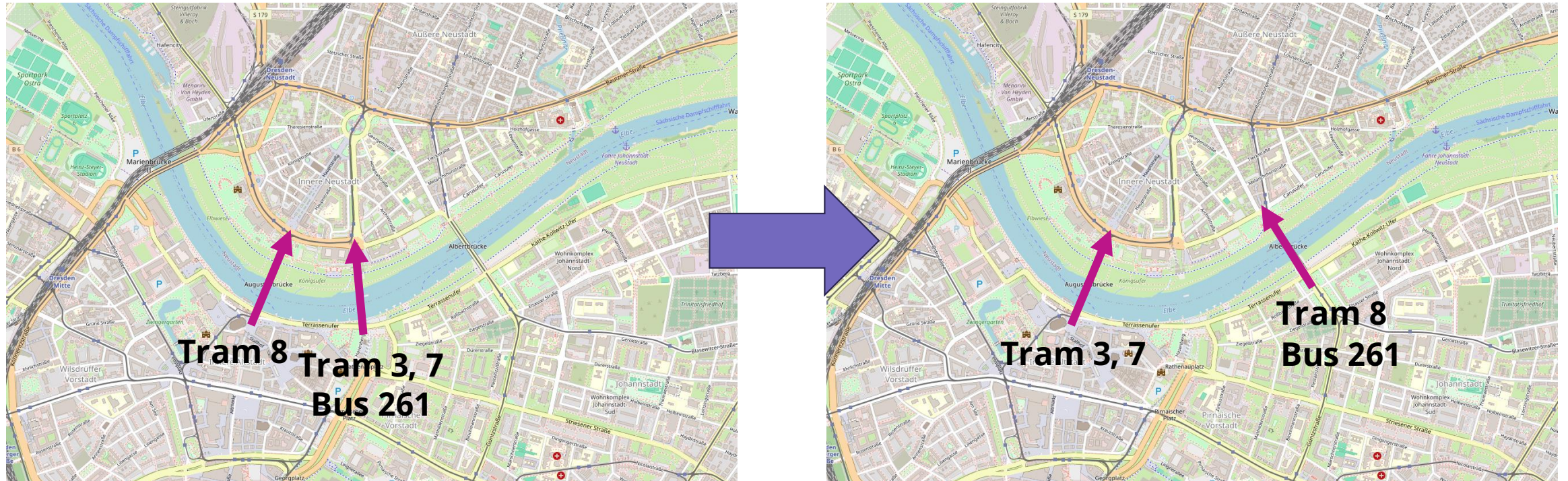
- Count data[^]: 65886
- Simulation: 103660
- Relative difference: +57.3%

*: For bridges included in the automated count data only.

[^]: Based on automated count based on various sensor data.



Update PT schedule based on the reality



The other PT lines remain untouched

Motorized vehicle (Kfz)

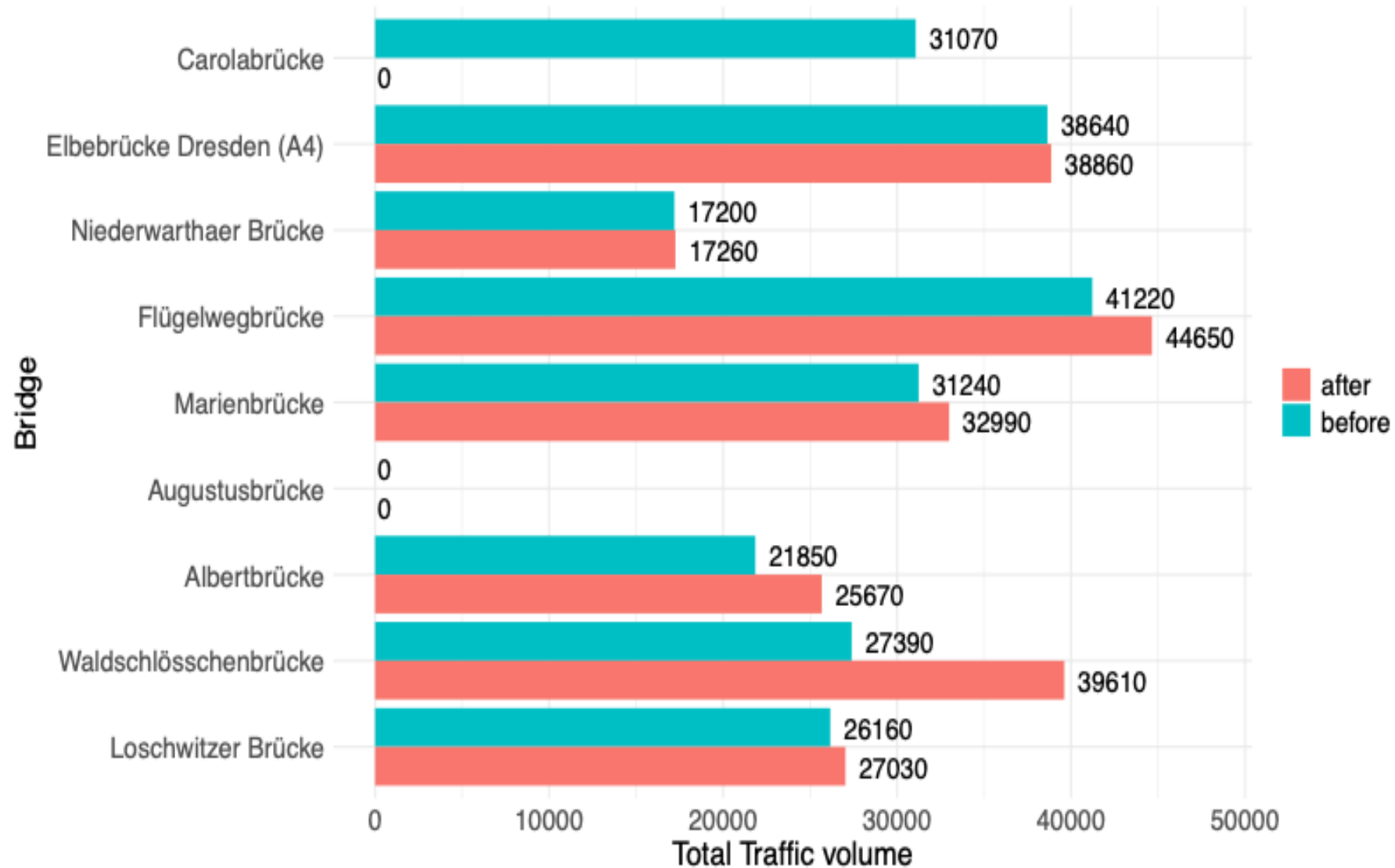
Prediction

The impact of the Carola bridge from simulation results:

Total count on all bridges

- Before: 234770
- After: 226070
- Difference: -8700 (-3.7%)

Impact of Carola bridge: total motorized vehicle (Kfz) volumes on the bridges



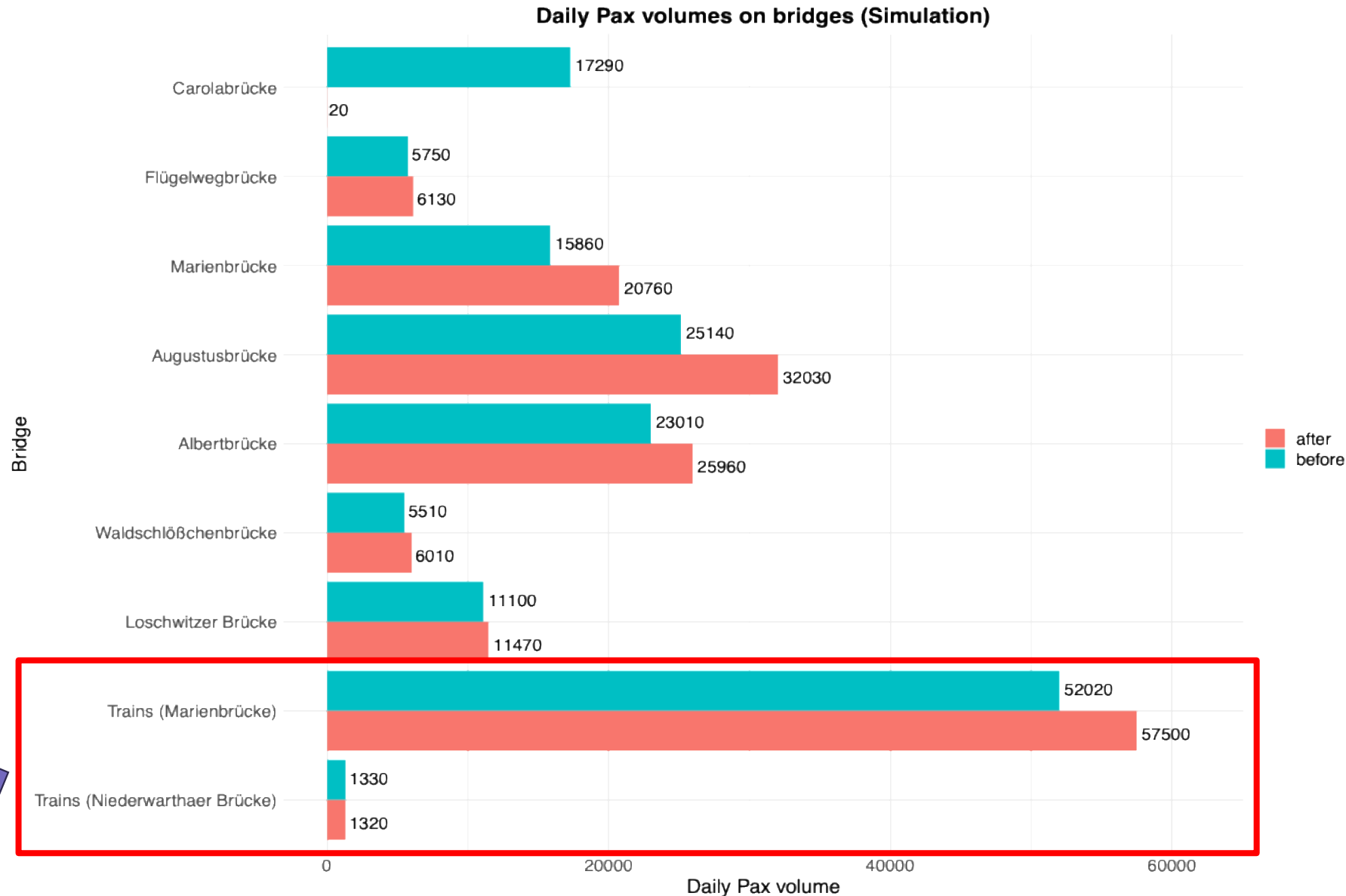
PT passenger traffic

Prediction

The impact of the Carola bridge from simulation results:

Total count on all bridges

- Before: 157010
- After: 161200
- Difference: +4190 (+2.7%)



Extra insight into the system provided by ABM

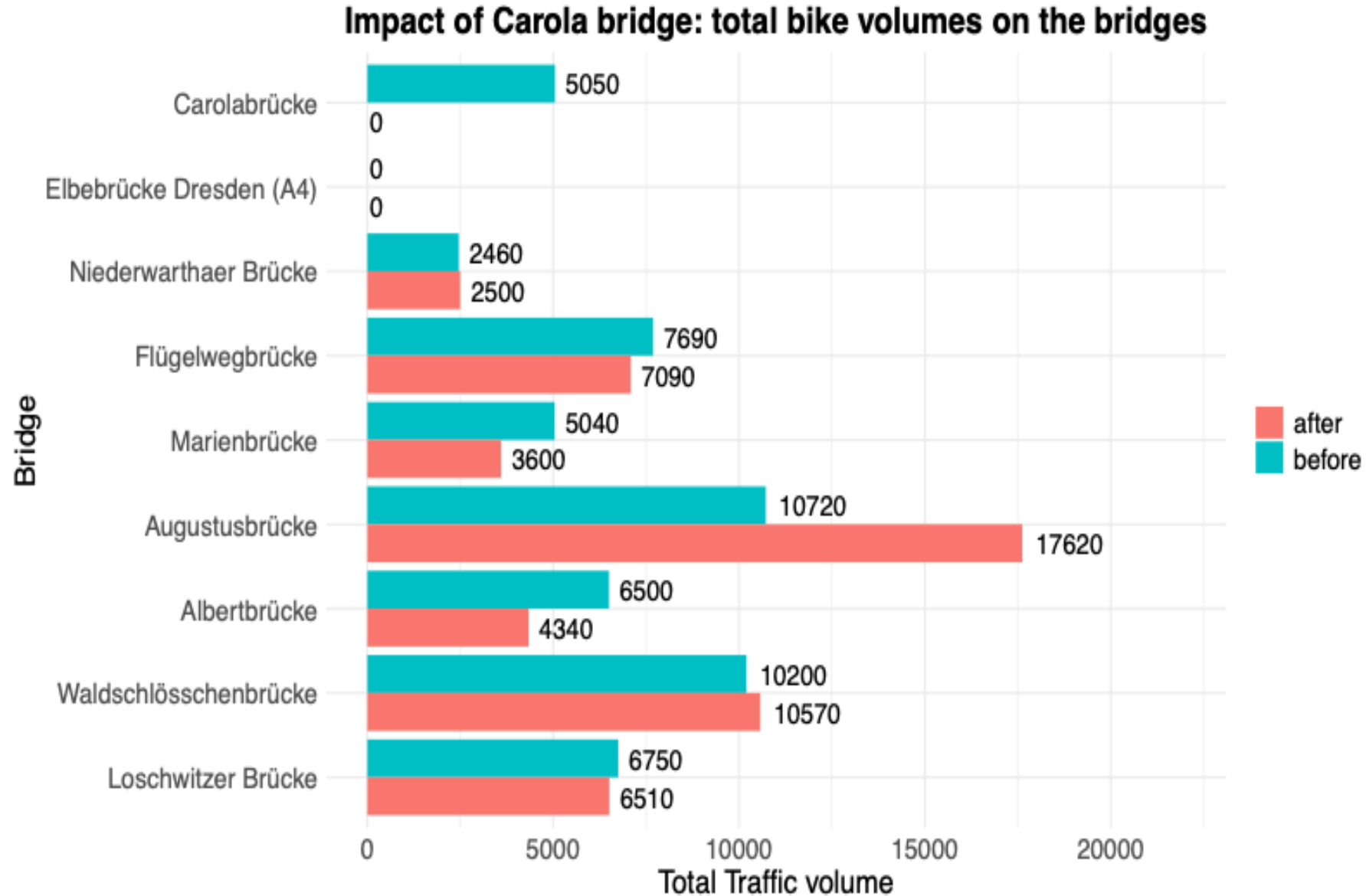
Bike traffic

Prediction

Before the collapse of the Carola bridge.

Total count on all bridges:

- Before: 54410
- After: 52230
- Difference: -2180 (-4.0%)



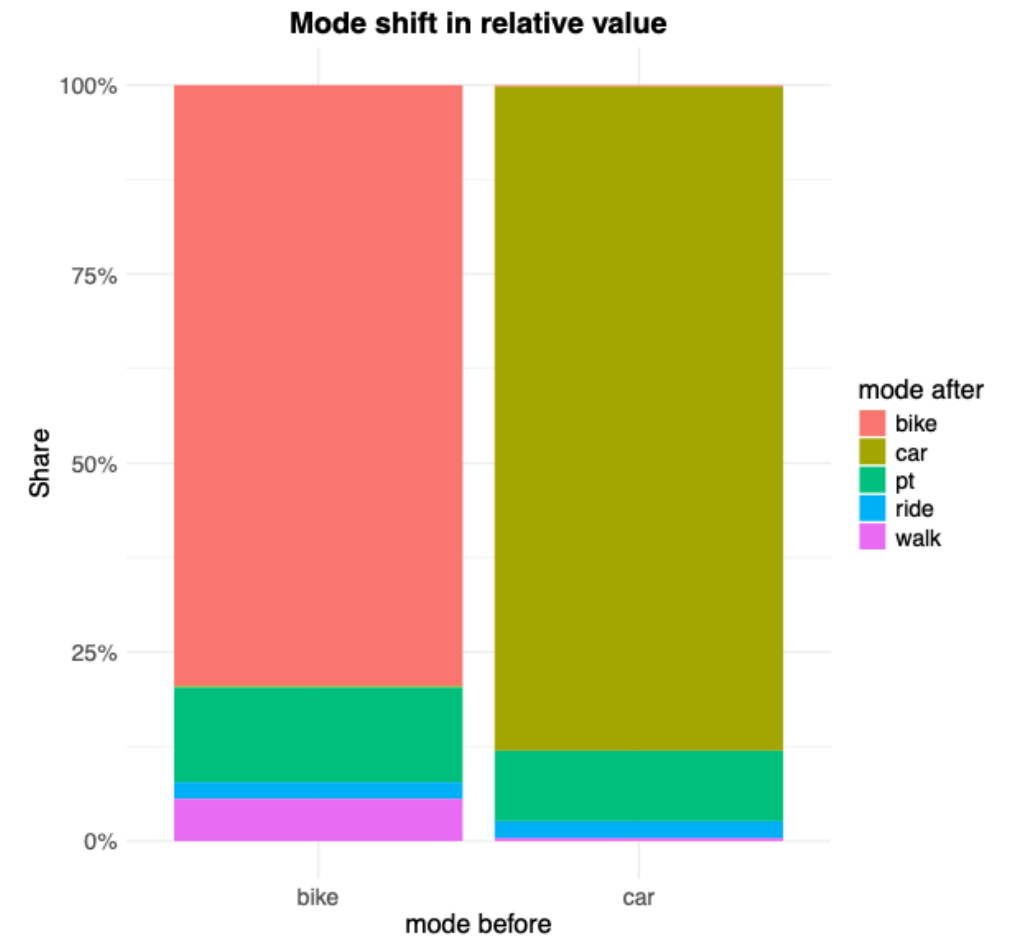
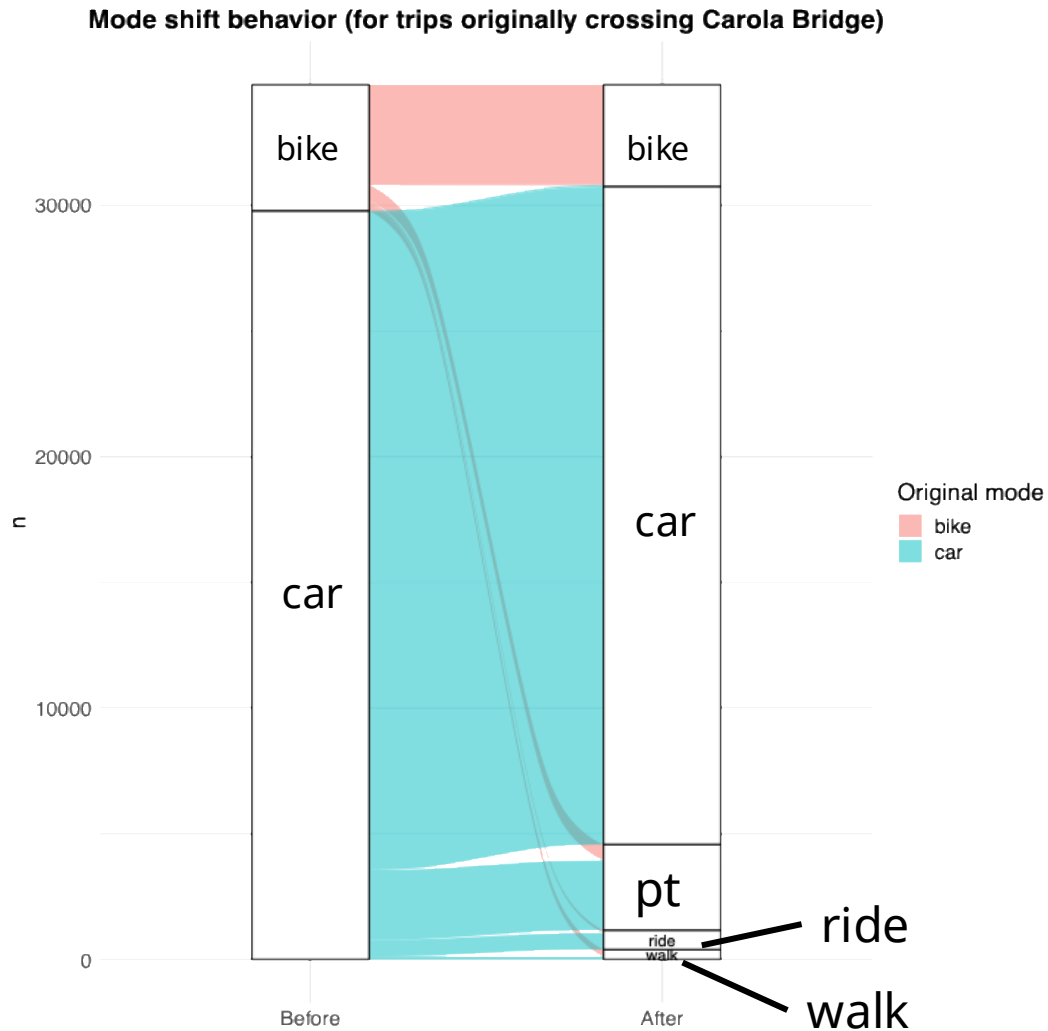
River crossing trips all together

Mode	Before	After	Difference
Car	234770	226070	-8700
Bike	54410	52230	-2180
PT	157010	161200	+4190
Ride*	/	/	+4470
Walk*	/	/	+2240

*: Only including the incoming mode shift from bike and car. The other mode shift should be trivial

Part II: agent-level results

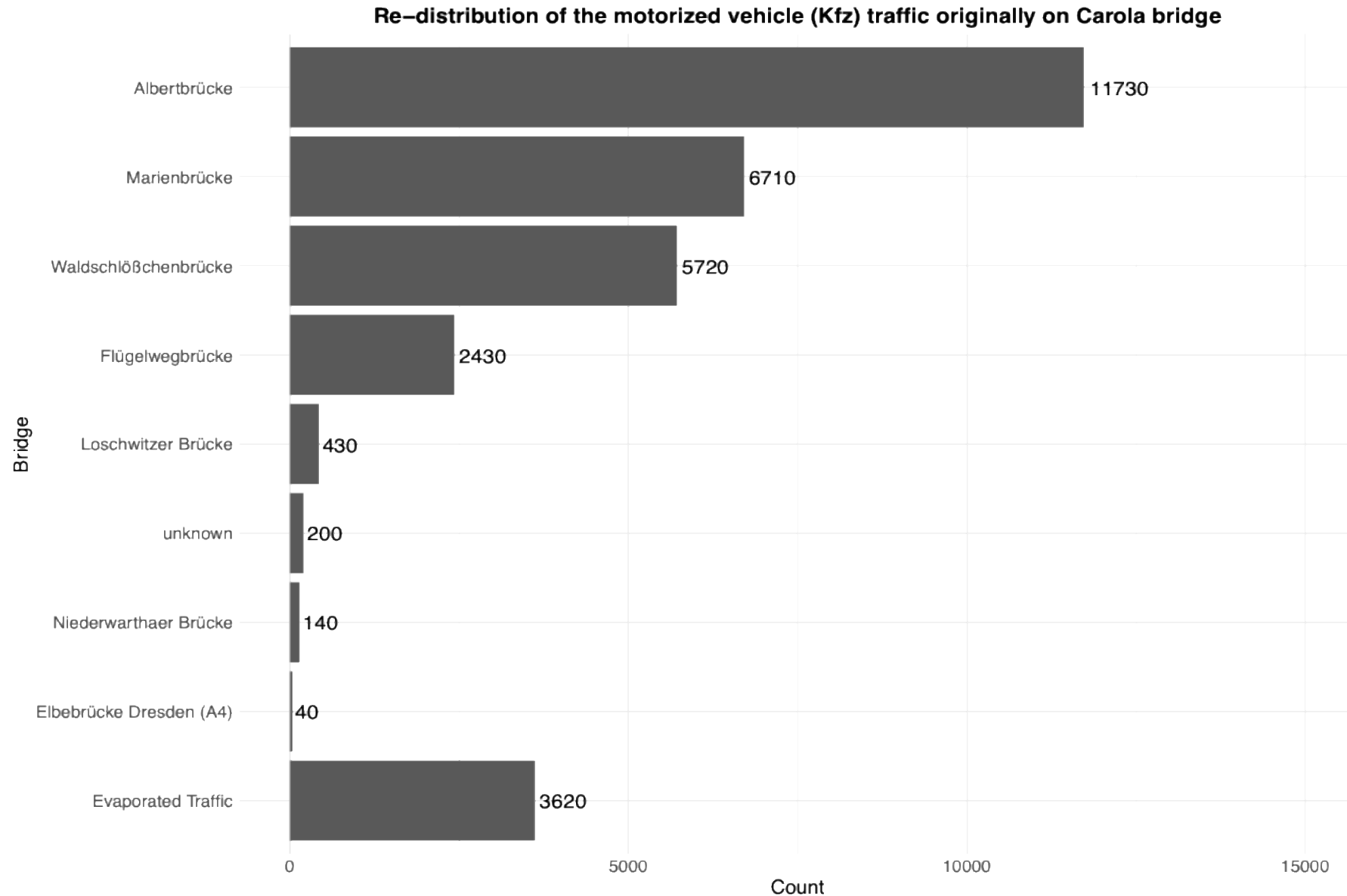
Mode shift behavior of the previous Carola bridge users



Motorized vehicle (Kfz)

Prediction

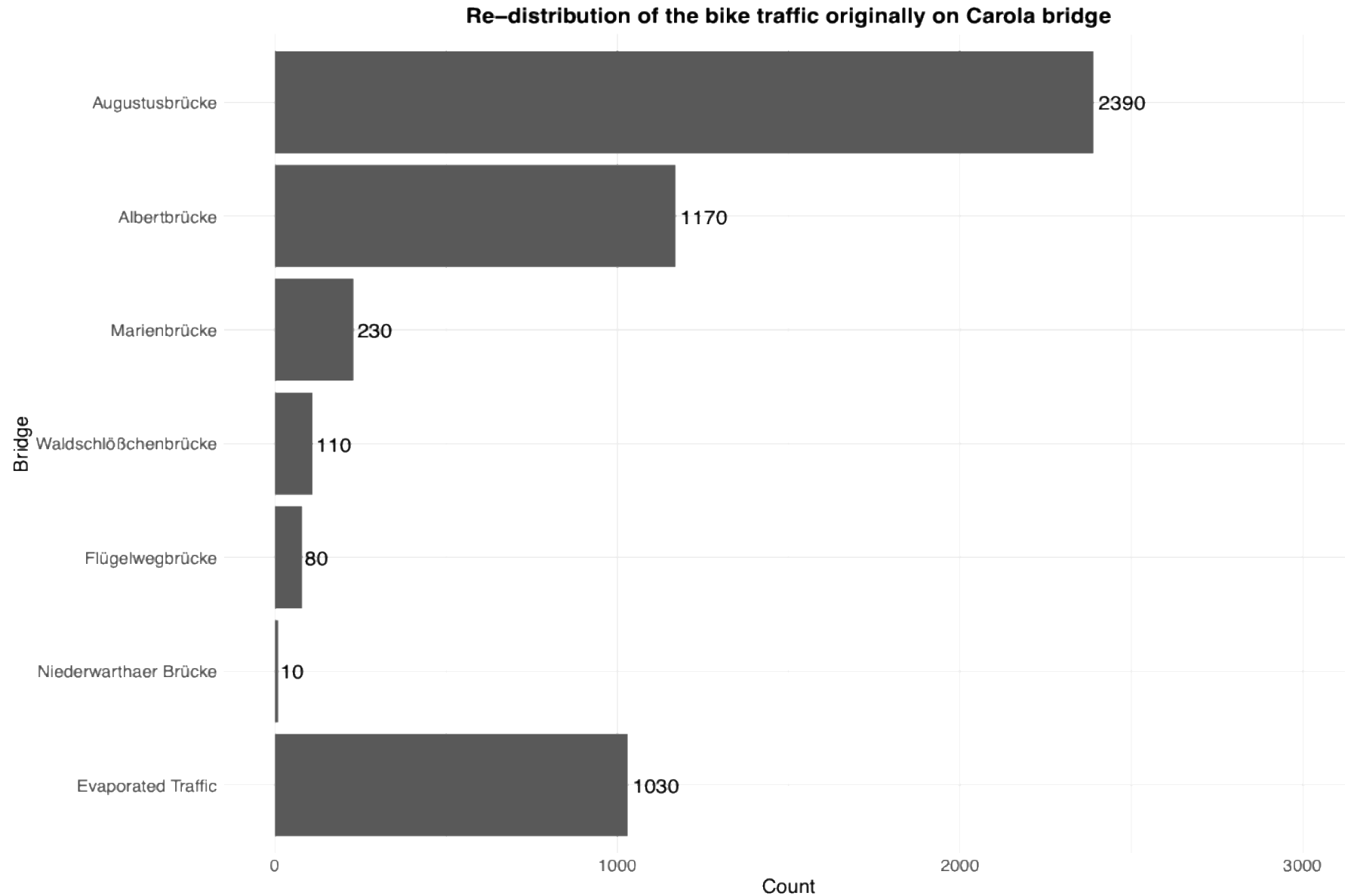
Traffic re-distribution of the previous Carola bridge trips.



Bike traffic

Prediction

Traffic re-distribution of the previous Carola bridge trips.



Additional river-crossing trips by ride and walk

Prediction

	to Ride	to Walk
From Car	2580	570
From Bike	1890	1670
Sum	4470	2240

*: Here we include also the traffic trips that re-route to other bridges outside the analysis area

Trips travel time analysis

System overall: +0.96% (for residents in Dresden: +1.16%)

- Previous Carola Bridge trips (car + bike): +10.19%
- DD Bridges trips (car + bike): +3.48%
- Trips **without** crossing the Elbe river within Dresden (car + bike): +0.47%

* car: motorized vehicle (Kfz), also includes trucks

Open discussion

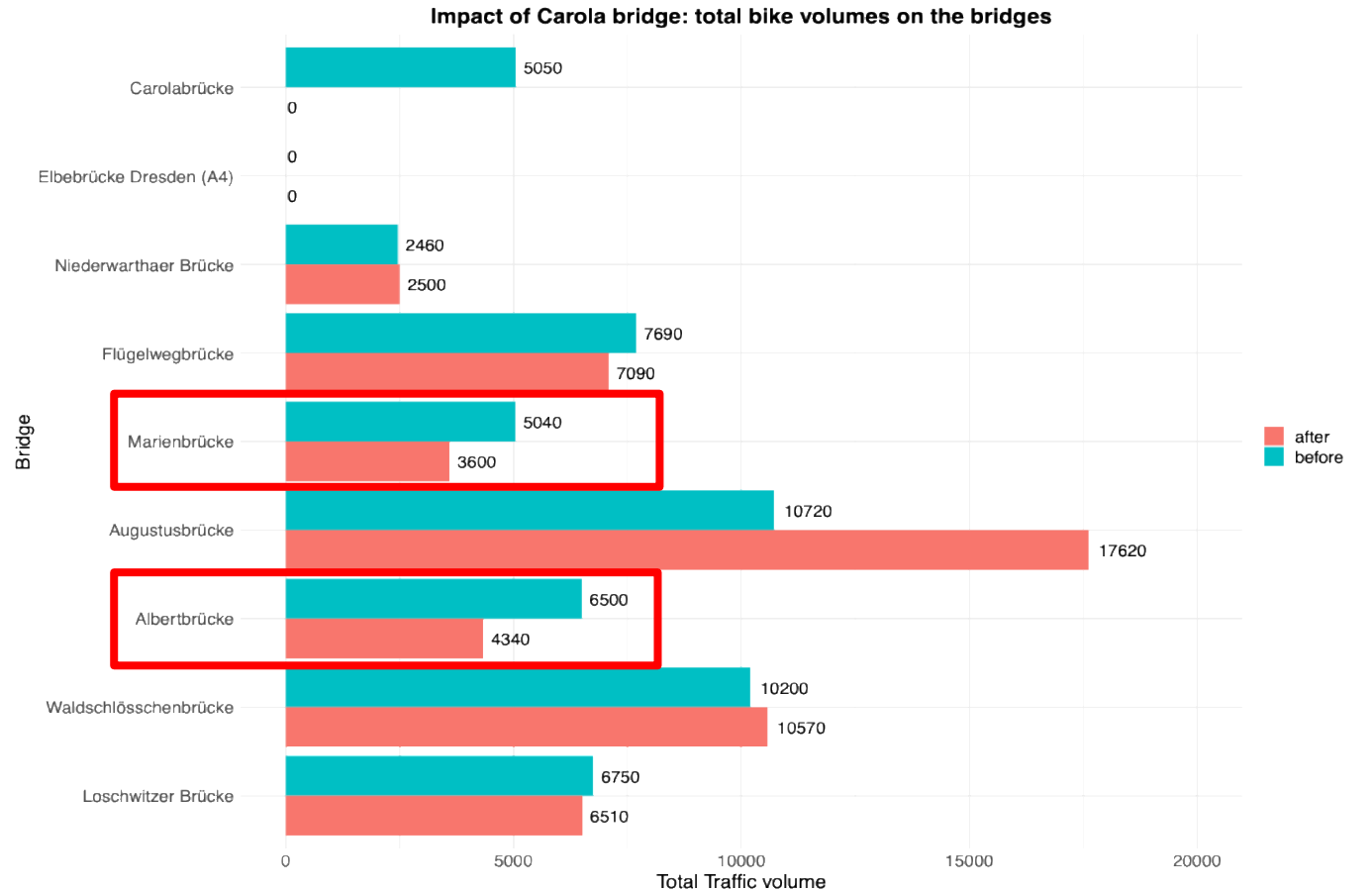
- The bike behavior
- Matching the car traffic count before and after the collapse: the waterbed effect
- Impact of activity and destination changes in the reality
- Reconstruction of the Carola bridge: simulation of various setups?

The bike behavior

Bikes is congested with car traffic on the Marien bridge and Albert bridge.

So the bike traffic actually decreased on those two bridges!!!

Not the case in the reality!



Solution

A. Enable bike seeping, and set bike PCU to 0

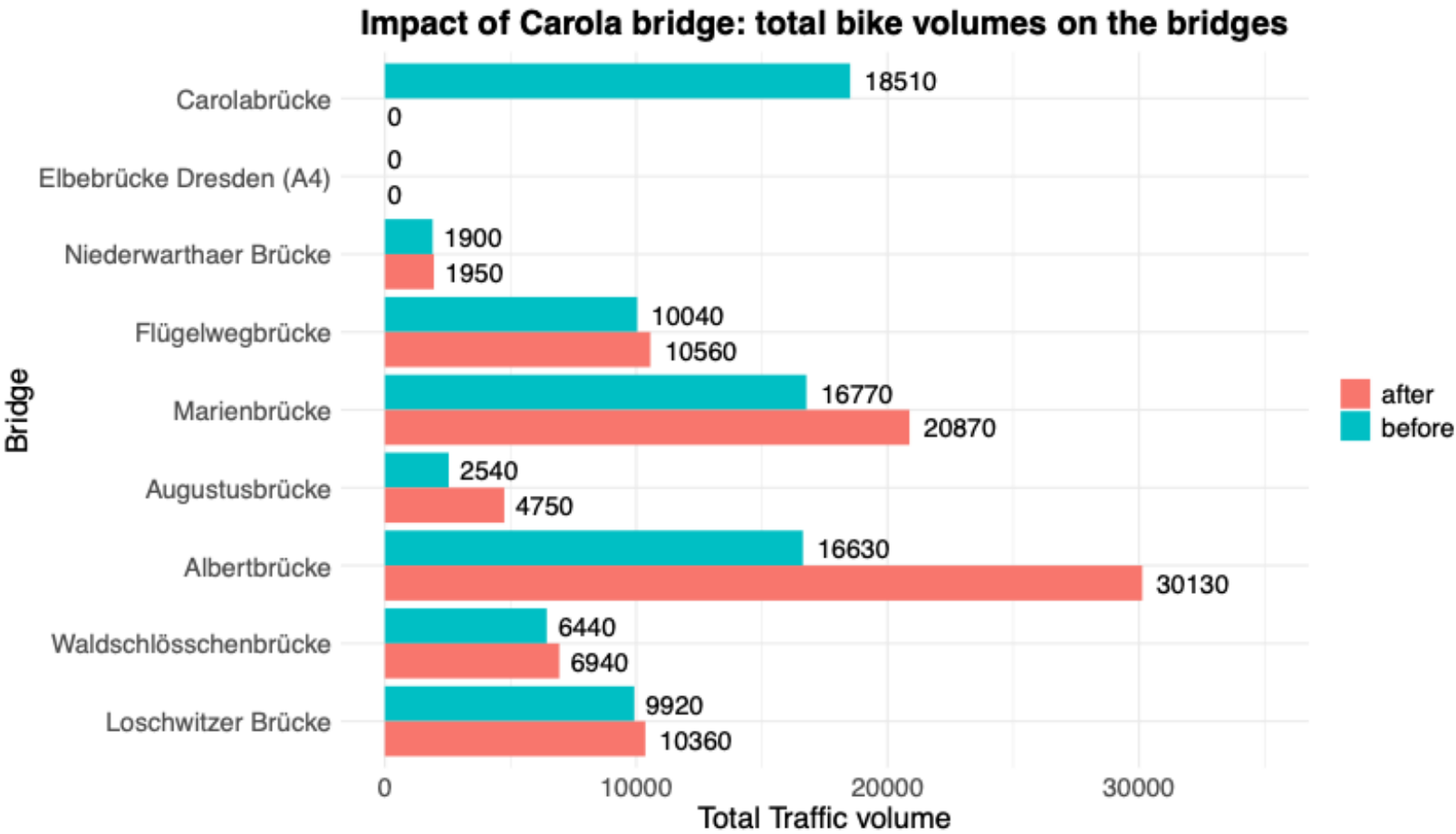
- **Advantage:** No need to change the network, solves the problem with minimum changes
- **Disadvantage:** longer simulation time; the function is not commonly used, may have hidden bugs.

B. Bike running on a separate network

- **Advantage:** does not require experimental module; **we can set different free speed and even travel cost for bike links (independent of the car links)**
- **Disadvantage:** the network becomes a little more bulky; visualization with Simwrapper/VIA may need some adaptations.

The Dresden model with separated bike network

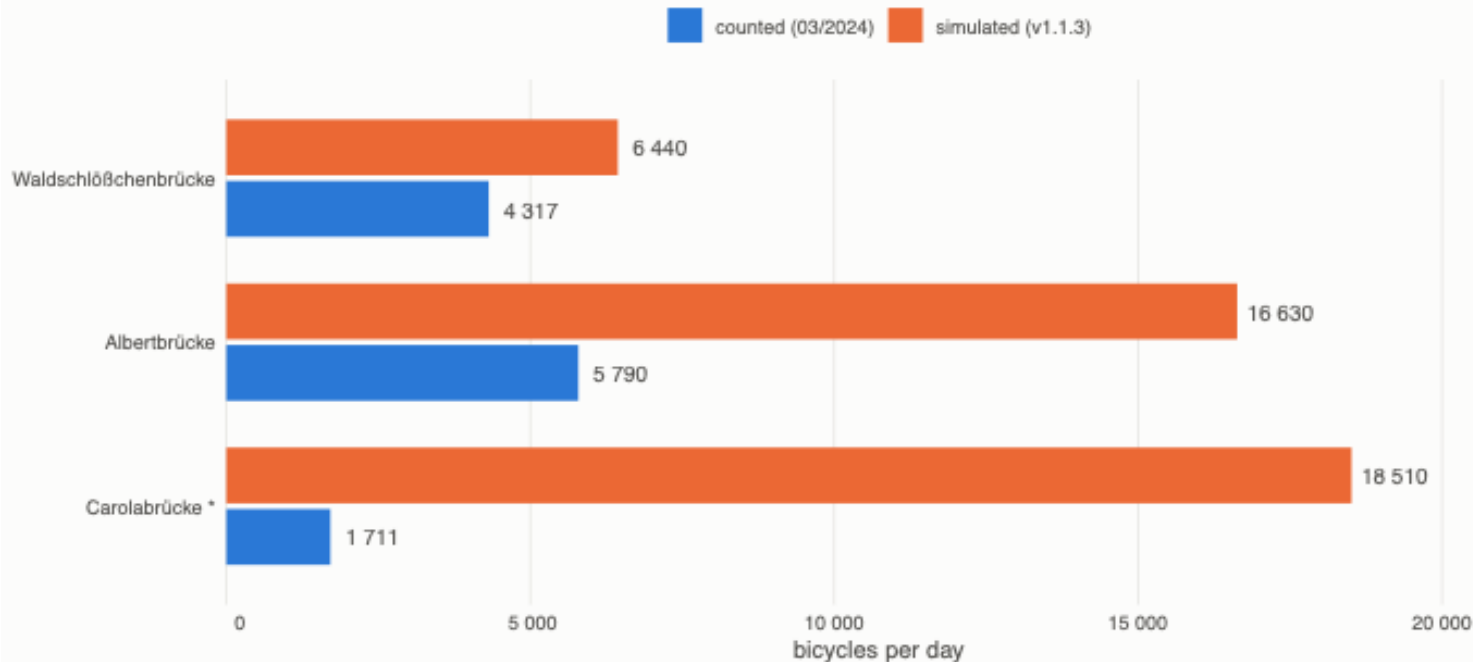
Redistribution of
bike traffic



The Dresden model v1.1.3 with separated bike network

Elbe bridge cycling volumes: counted vs simulated

Bicycles per normal working day, both directions. Only the three bridges with a counter on the deck are shown – Marienbrücke and Loschwitzer Brücke were not yet counting in 03/2024, the Flügelwegbrücke counter is on the Elberadweg beneath the bridge, and the Augustusbrücke has none. Run: dresden v1.1.3 base case, 10 %



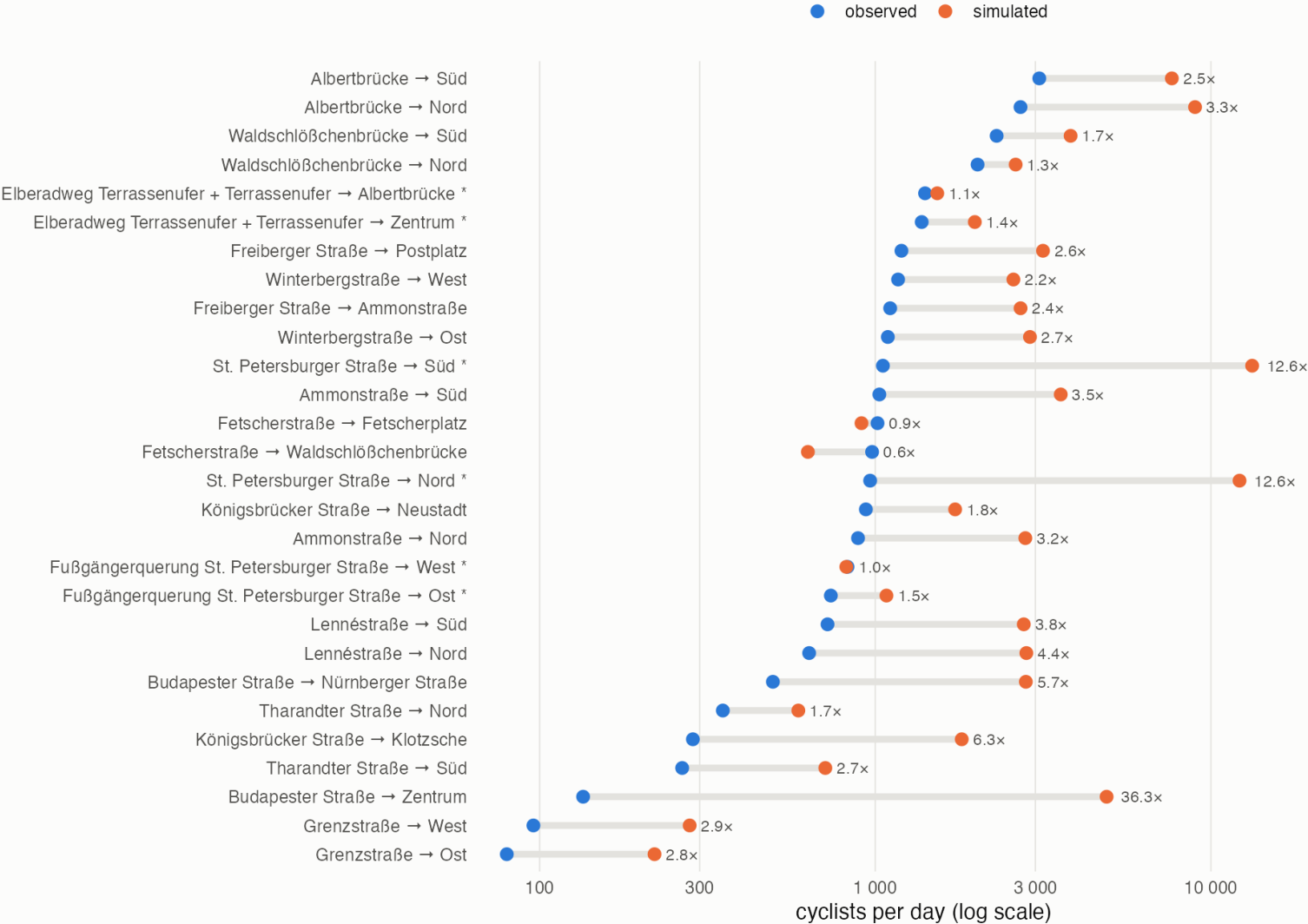
* Carolabrücke: the counter measures only the Altstadt-bound side, and never measured a March – the value is the August 2024 count scaled to a March equivalent and then grossed up to a full crossing. It is an estimate twice over.

Too much bike counts!!!

https://stadtplan.dresden.de/?bm=VECTOR&th=STA_FAHRRADZAEHLST&pos-box=408383.5%7C5652447%7C416048.5%7C5658897;25833&pos-mark=false

Simulated bicycle volumes run well above the counting stations

dresden-v1.1.3 base case, 10 %, it. 500 vs. the mean normal working day of 2024-03 - 28 cross-sections, each direction counted separately



Number beside each pair: simulated / observed. * = the link matching is approximate - two counting stations share one network link, or the nearest bike link is more than 20 m from the station. Counts: Landeshauptstadt Dresden, Dauerzaehlstellen Radverkehr (Open Data). Simulation: MATSim count-comparison, mode bike, scaled up from the 10 % sample.

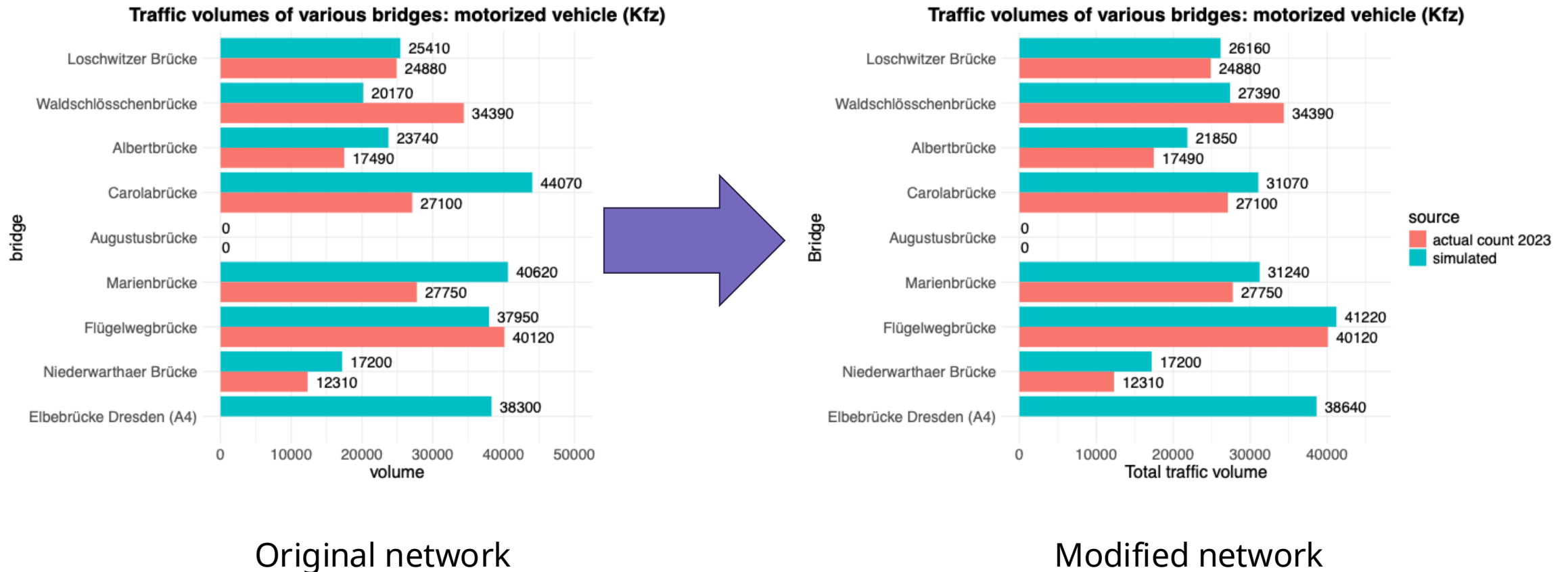
Speculation and potential solutions

The automated counting station tends to **underestimate** the actual count → We are already requesting the **manual count data** from the city

The time of the year may also play a role (e.g., currently March 2024 is used)

The improved bike-contrib should hopefully also address the problem

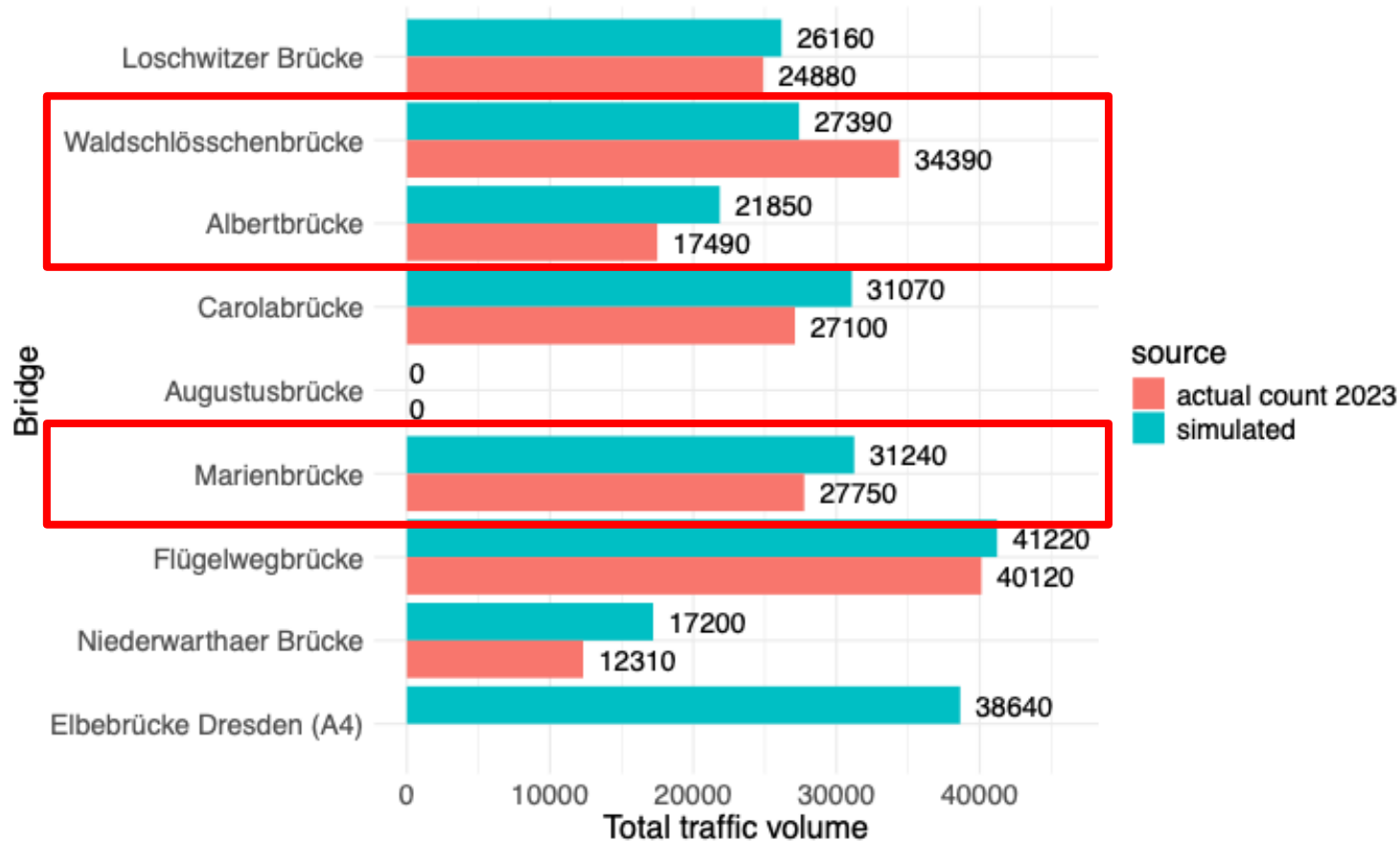
Matching the bridge traffic count in the base case



Matching the traffic count data on the bridges

The Dresden model v1.1 with modified network

Traffic volumes of various bridges: motorized vehicle (Kfz)



Evaluate and update the capacity of the bridges.

Reduce capacity of:

Carola bridge

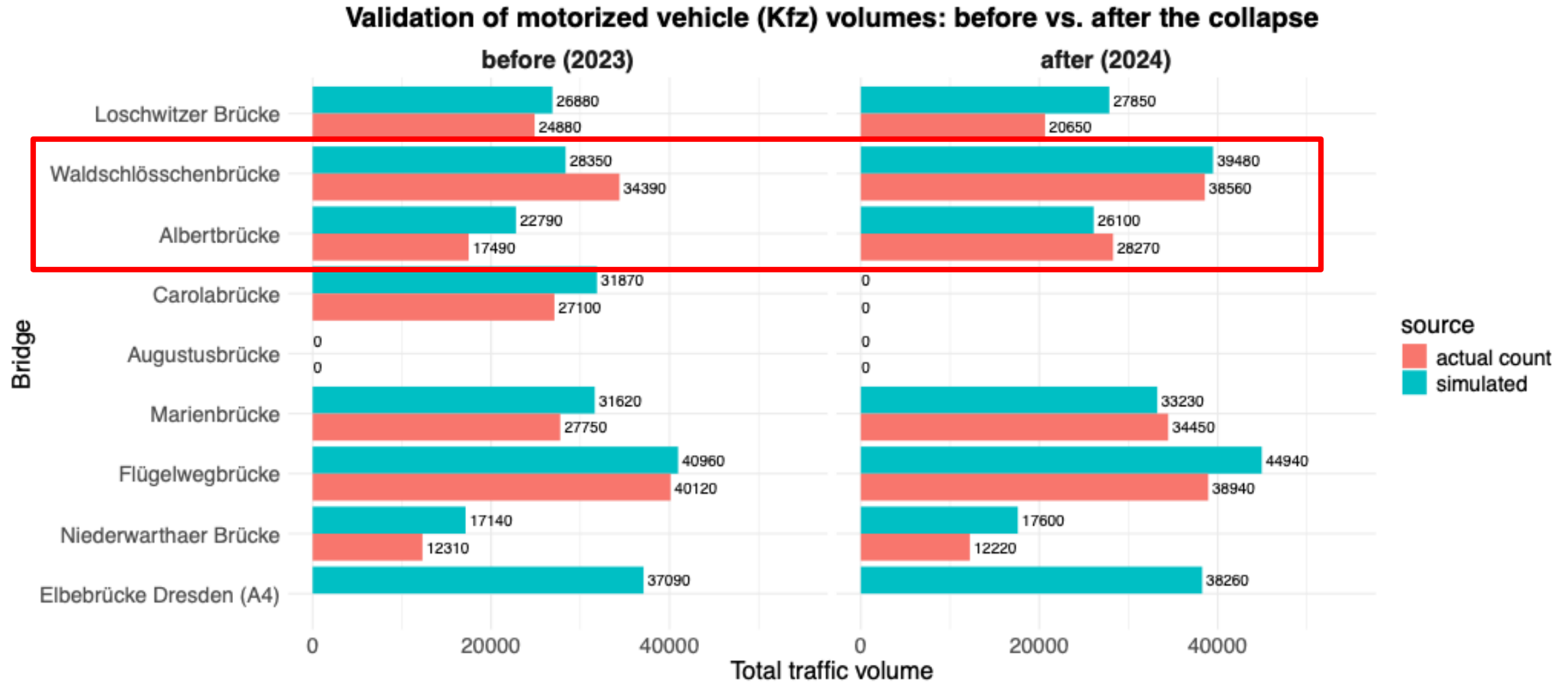
Marien bridge

Albert bridge

Increase the capacity and freespeed of:

Waldschlosschen bridge

After the collapse



The waterbed effect

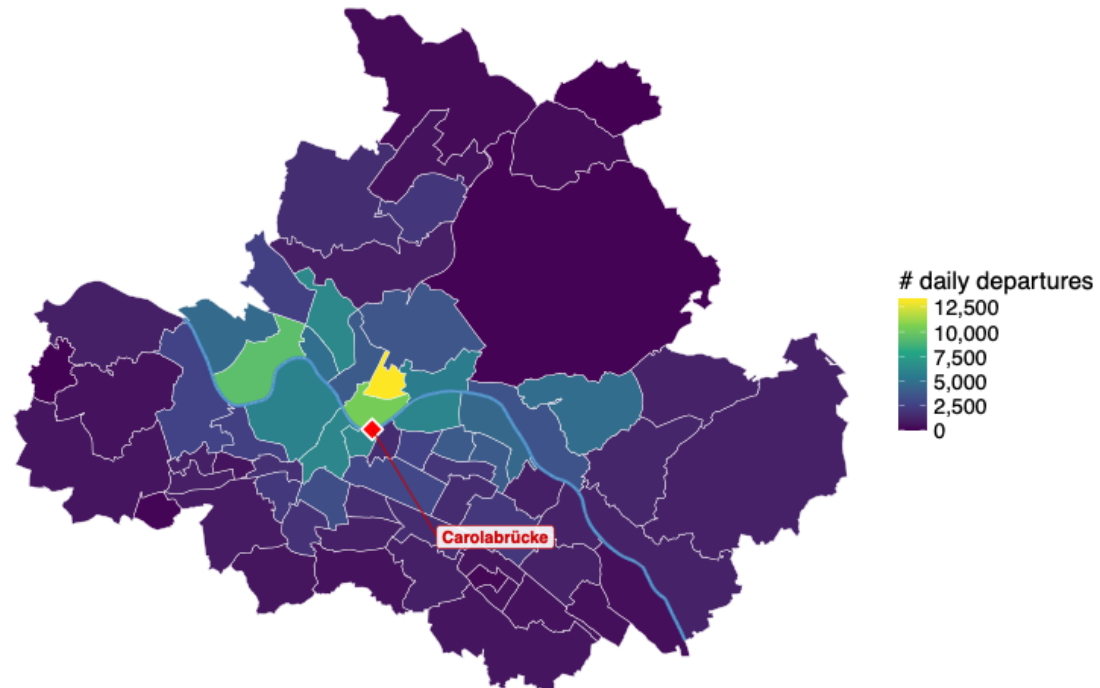
By reducing the capacity of the Albert bridge, we can reduce the mismatch in the car traffic count in the base case (i.e., before the collapse of the Carola bridge). However, we underestimate the traffic after the collapse of the Carola bridge.

If we do not reduce the capacity of the Albert bridge, we would overestimate the traffic on the Albert bridge in the base case.

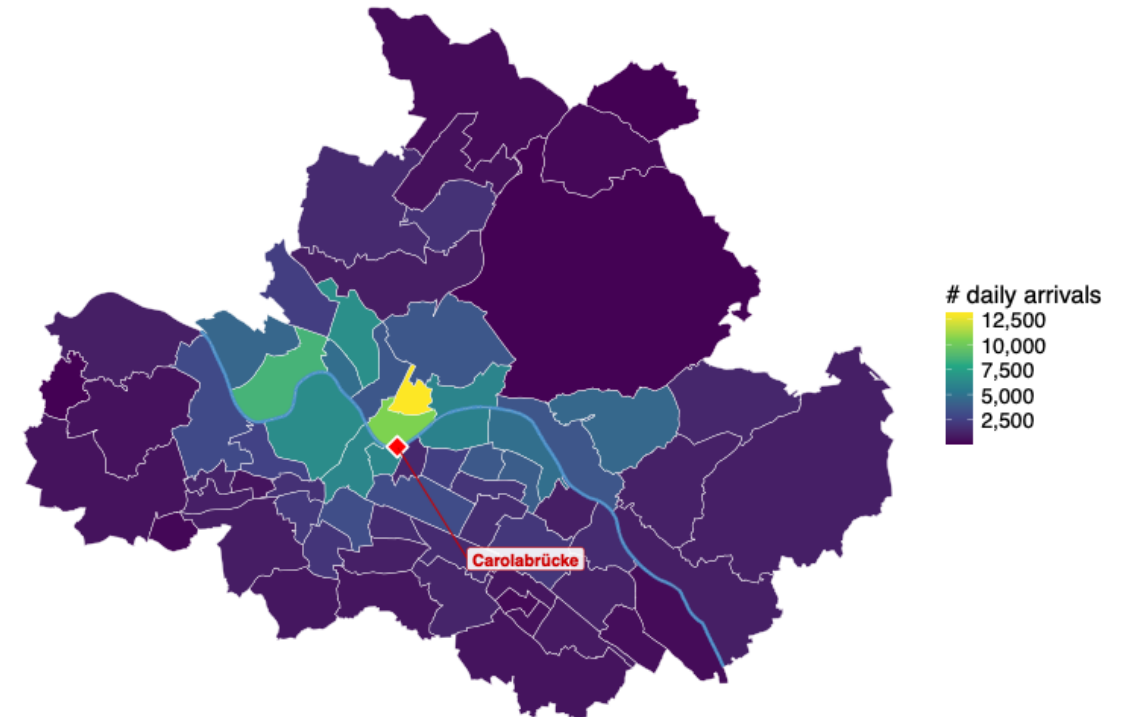
And we can also gain access to city-wide counting data by collaborating with other research group.

Impact of activity and destination changes in the reality

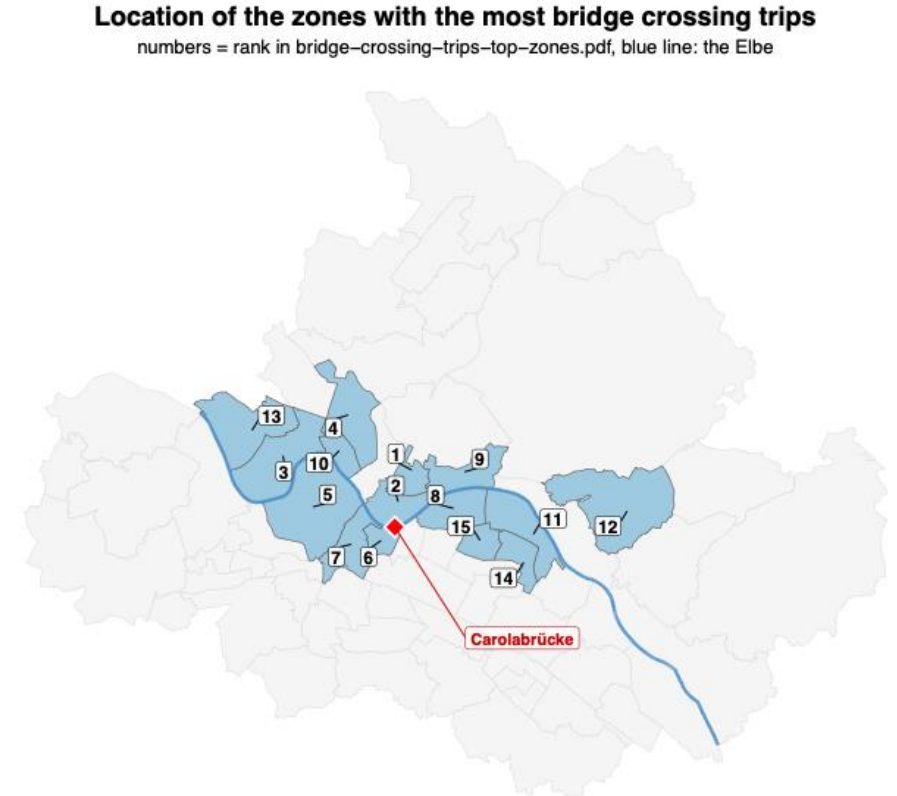
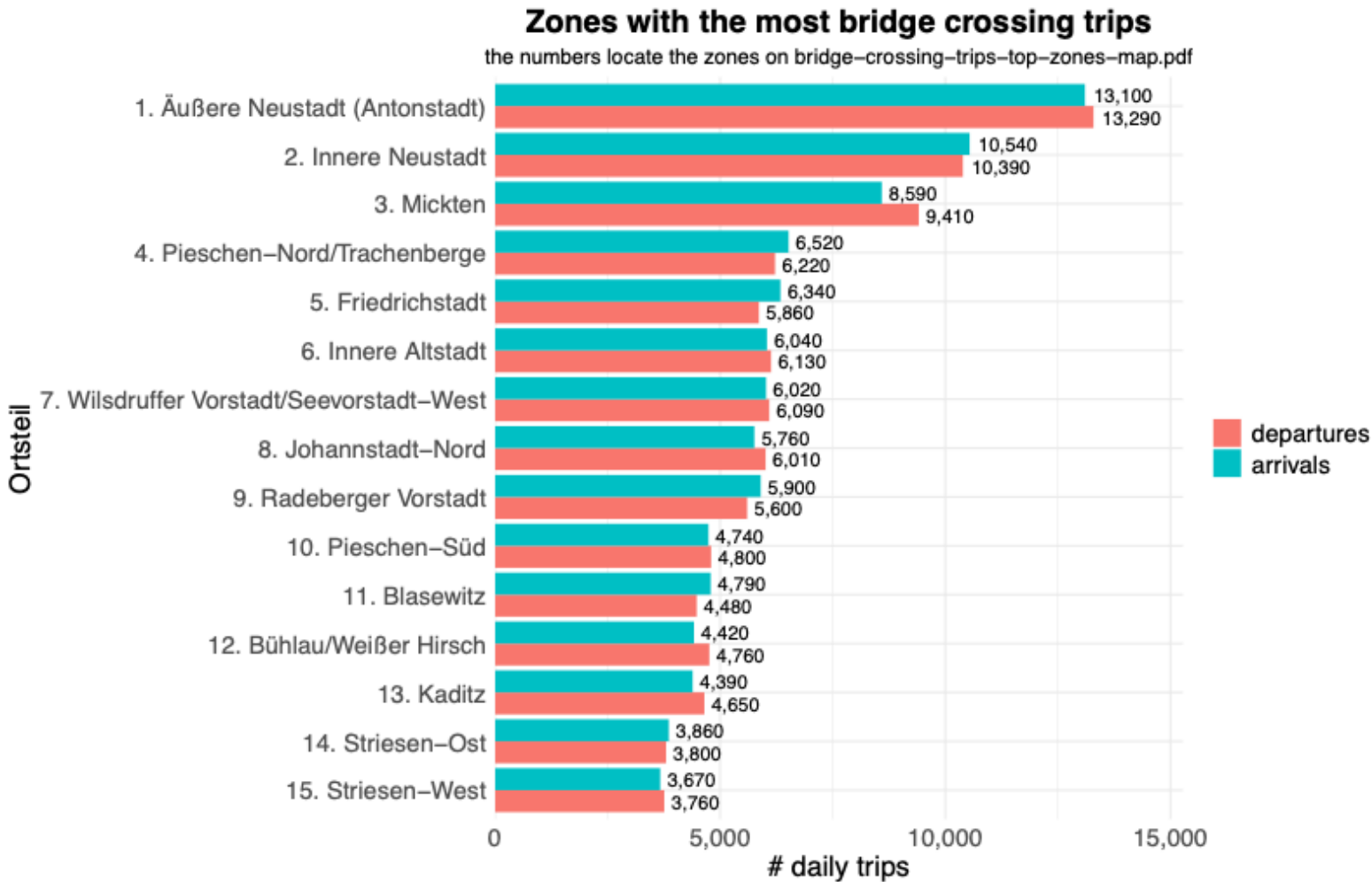
Spatial distribution of bridge crossing trips: departures
all trips crossing a Dresden Elbe bridge (car incl. truck and bike), blue line: the Elbe



Spatial distribution of bridge crossing trips: arrivals
all trips crossing a Dresden Elbe bridge (car incl. truck and bike), blue line: the Elbe



Impact of activity and destination changes in the reality



Reconstruction of the Carola bridge

The reconstruction of the Carola Bridge is now on the schedule of the city.

How could we use the Dresden Model to provide decision support for the transport planning?

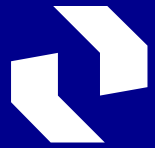
- How many lanes should there be (i.e., capacity limit of the link), and what would be the impact
- The speed limit and it's impact
- Re-route the Bundesstrasse B170 in the city?

Reference and acknowledgement

Acknowledgement:

MATSim Open Dresden Scenario used in the case study is prepared and made publicly available with partial funding from the German Research Foundation (DFG) – TRR: 408/1 2025 – Project number: 531327426. We also thank our colleagues from the Chair of Transport Systems Planning and Transport Telematics (VSP) at TU Berlin for setting up and maintaining the scenario.

The traffic count data used in this study are obtained from the VAMOS system, the strategic traffic-management platform for the state capital of Dresden, operated in cooperation with the City of Dresden.



Dresden
University of
Technology

Thank you very
much!



Backup slides

VAMOS Sensor Group

186 Double Loop Detectors

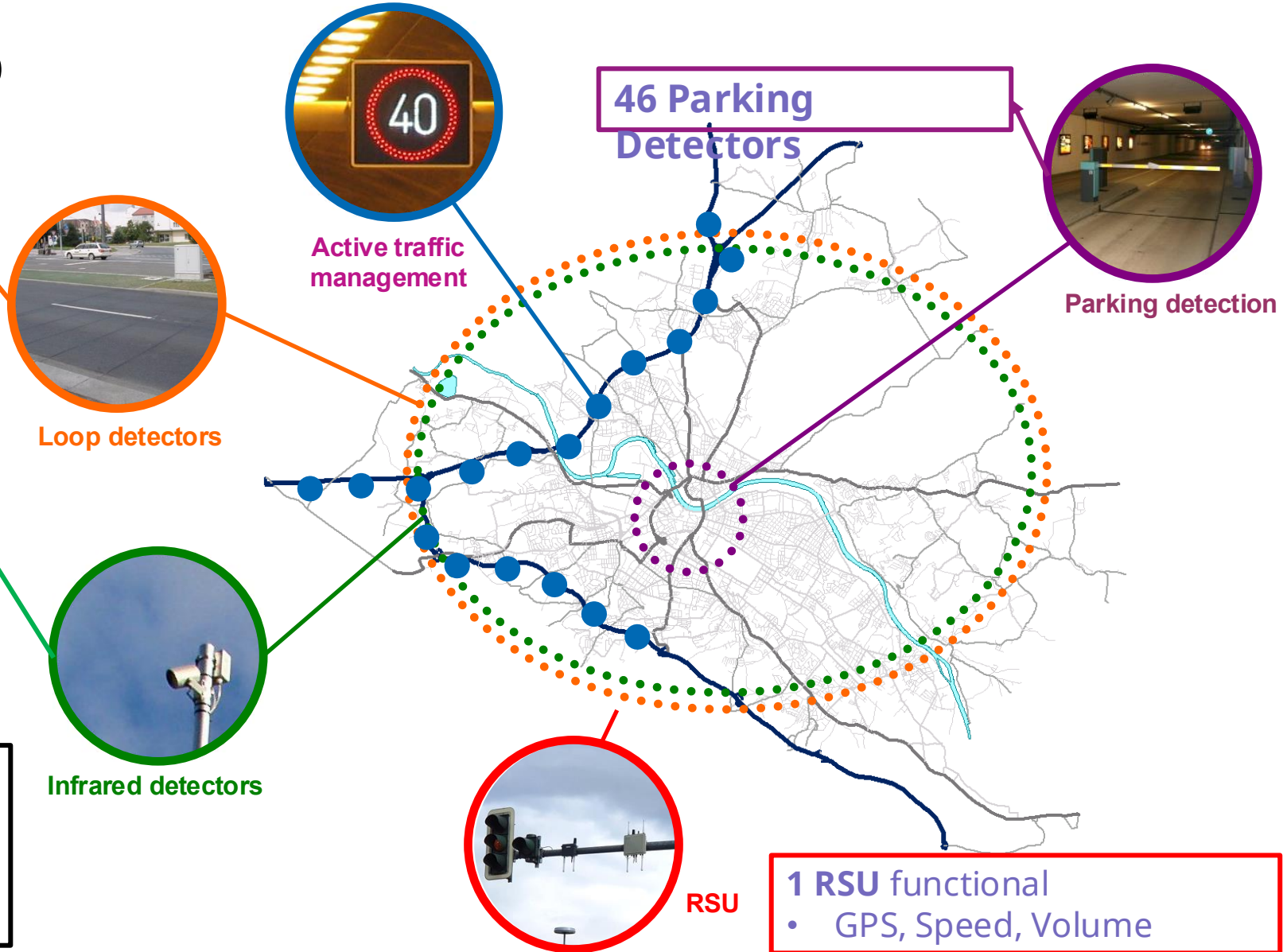
- Located at **61 sites**
- Measure **volume, speed, and vehicle classes**
- Data recorded every **1 minute**

160 TEU Detectors

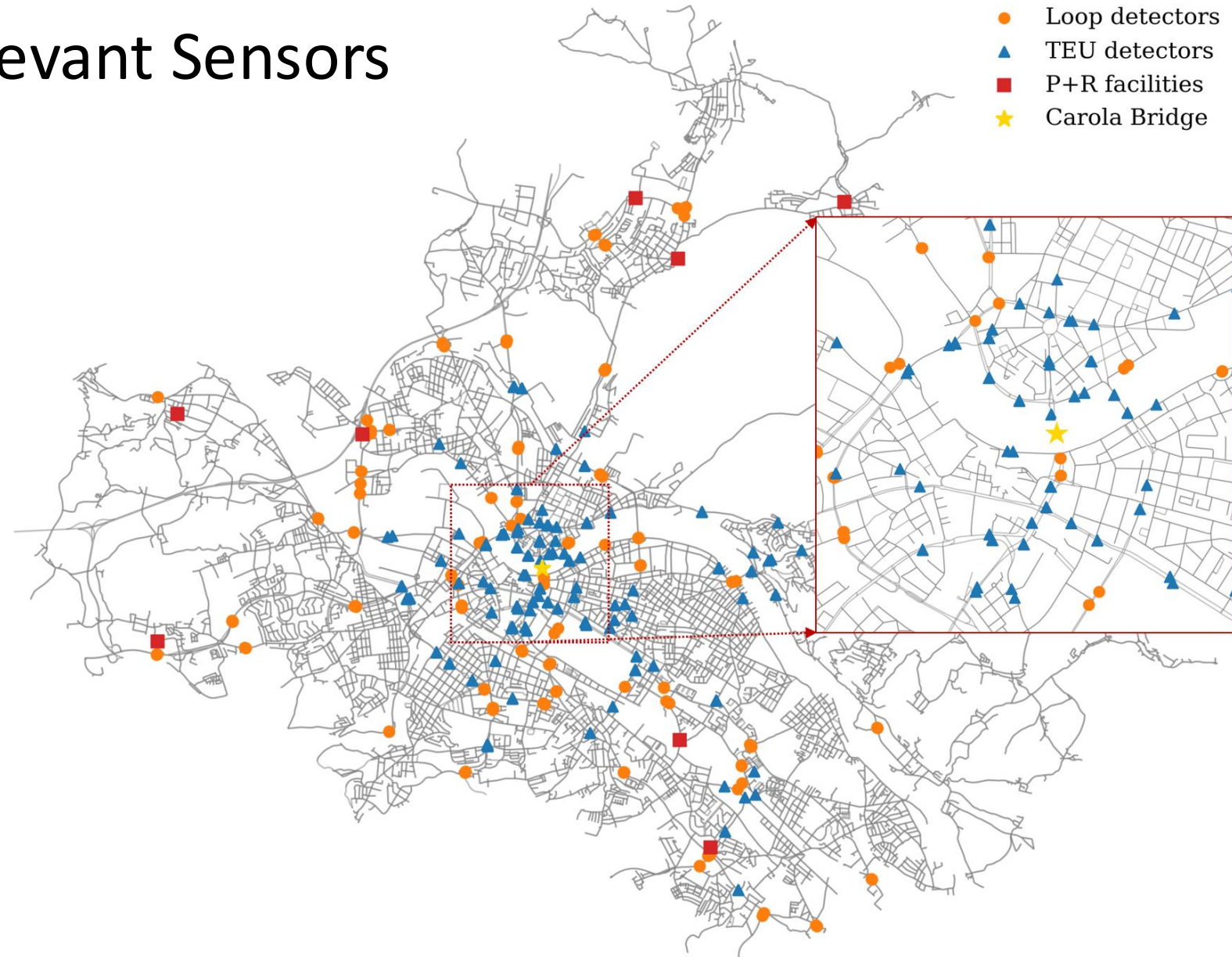
- Located at **93 sites**
- Measure **volume, speed and LOS**
- Data recorded every **3-5 minutes**

Historical Sensors

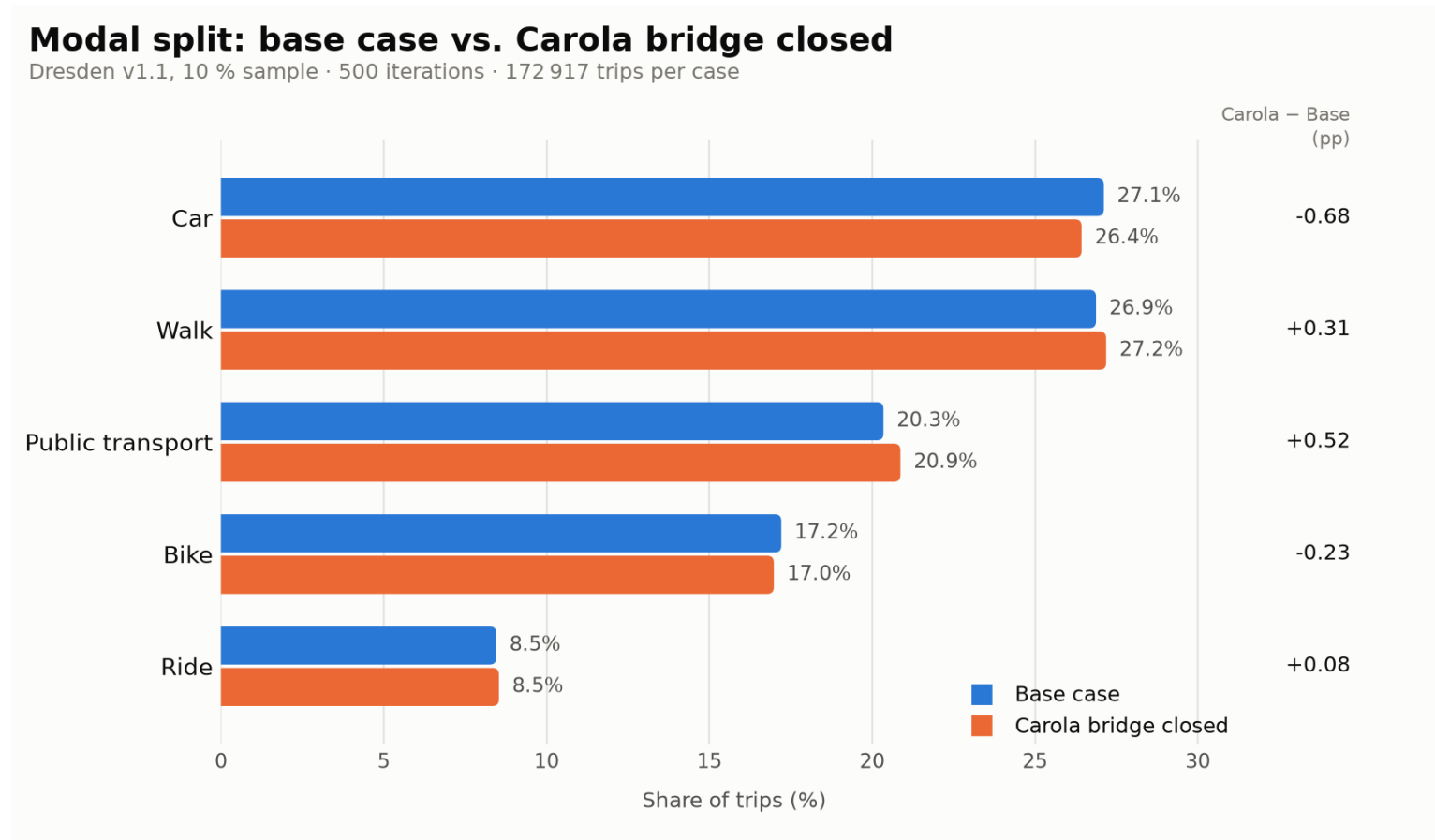
- **743 single loop** detectors at the traffic light
- **158 double loop** detectors in motorway



Locations of Relevant Sensors



Part I: System level results around the bridges



PT passenger traffic

PT Count data vs. Simulation

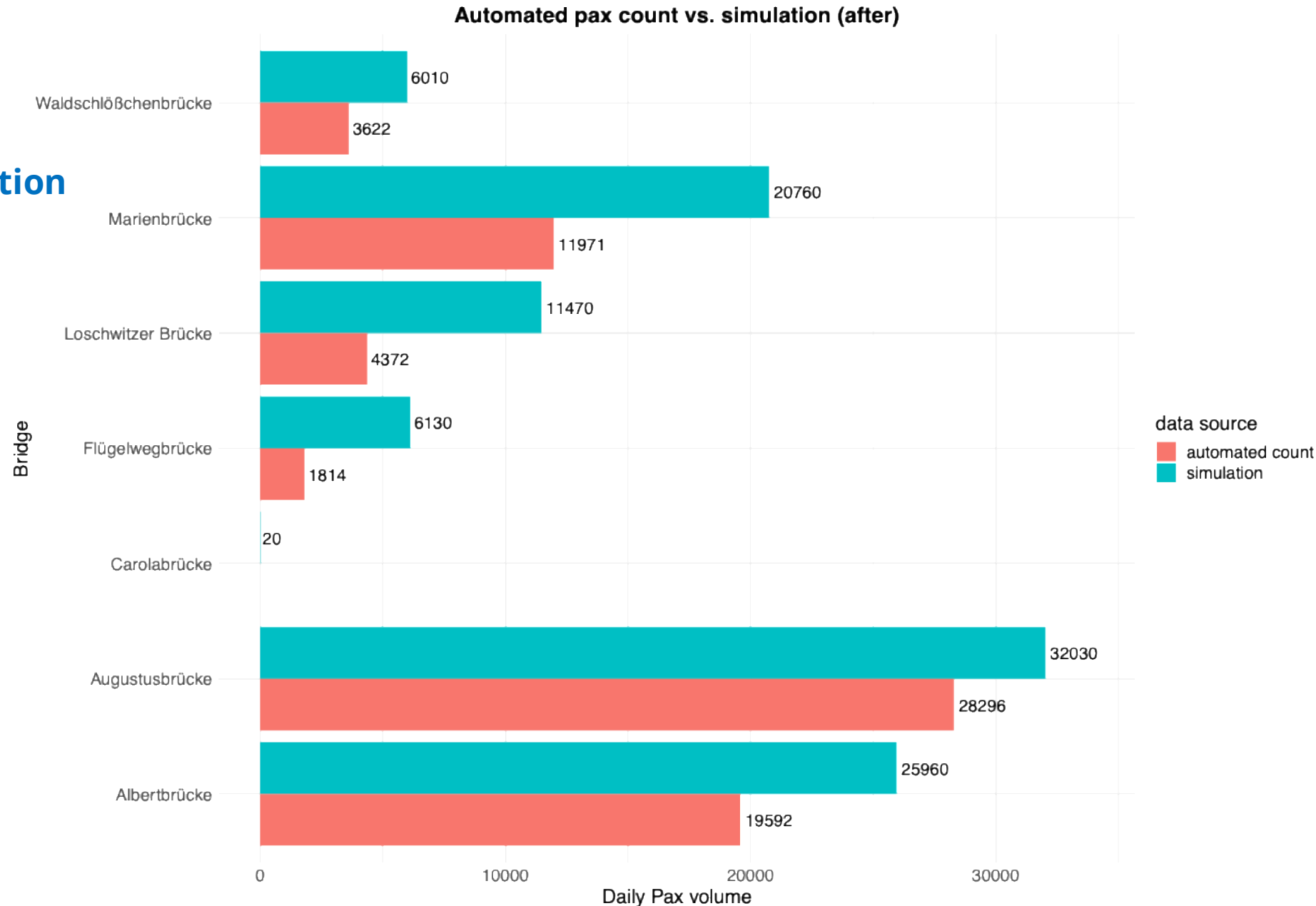
After the collapse of the Carola bridge.

Total PAX on all bridges*

- Count data[^]: 69667
- Simulation: 102380
- Relative difference: +47.0%

*: For bridges included in the automated count data only.

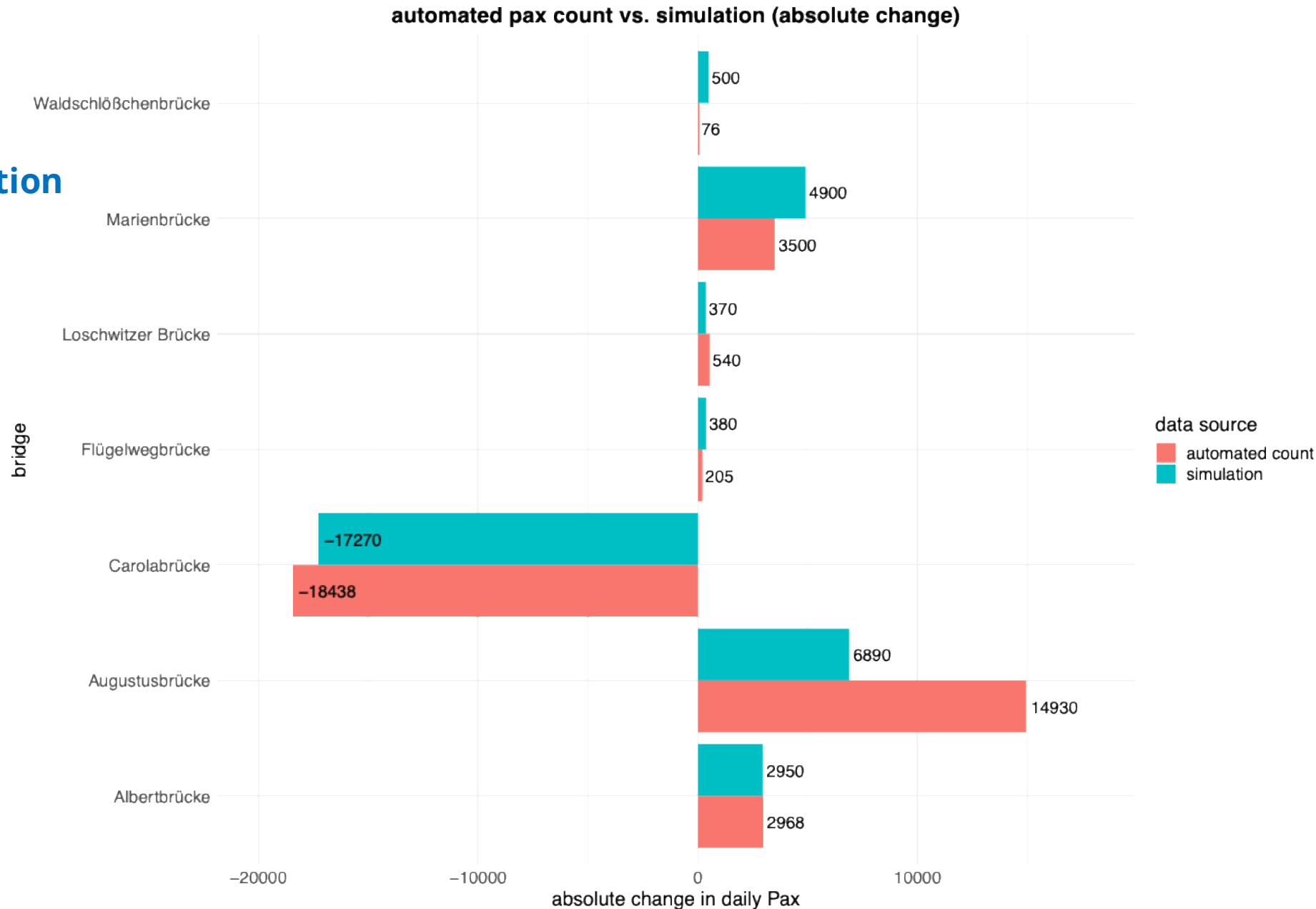
[^]: Based on automated count based on various sensor data.



PT passenger traffic

PT Count data vs. Simulation

Comparison of the trend
(i.e. the relative change):
Absolute value



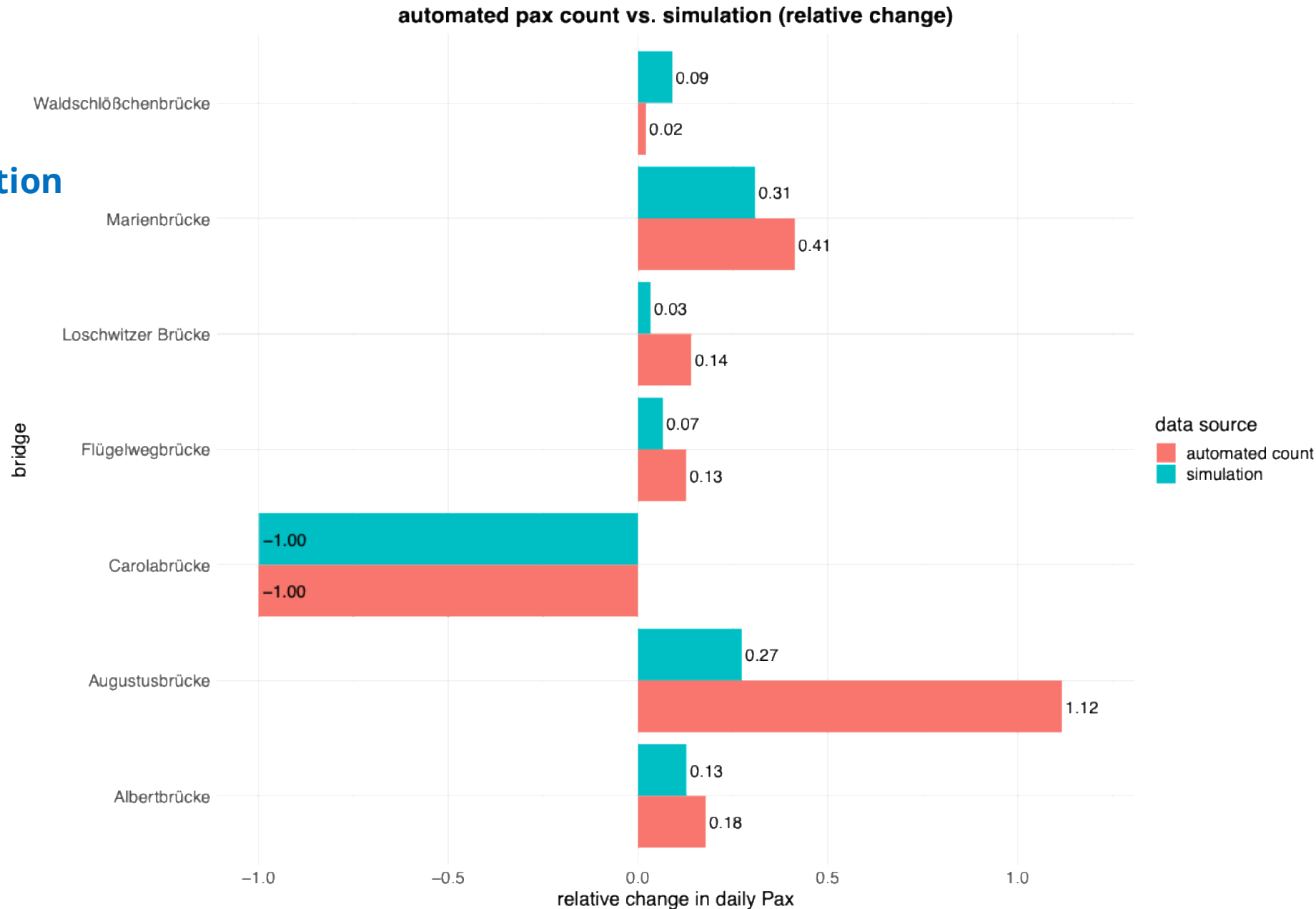
*: For bridges included in the automated count data only.

^: Based on automated count based on various sensor data.

PT passenger traffic

PT Count data vs. Simulation

Comparison of the trend
(i.e. the relative change):
Relative value



*: For bridges included in the automated count data only.

^: Based on automated count based on various sensor data.